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An exploration and exploitation of value cocreation-based machine learning framework for automated idea screening

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ABSTRACT

Idea screening in collaborative crowdsourcing communities poses significant challenges for firms. These challenges are primarily attributable to issues of prediction accuracy and information overload. The rapid expansion of idea pools generates a vast amount of data, making it difficult to effectively identify valuable ideas for new product development. This study introduces an interpretable framework for machine learning that integrates a novel exploration and exploitation perspective within the value cocreation model to enhance idea screening. The framework incorporates six theoretical dimensions of the exploration and exploitation of value cocreation (EEVC): the exploration and exploitation of digital resources, direct interactions, and ideas and their comments. Our evaluation reveals that the EEVC-based idea-screening system significantly outperforms the traditional 3Cs model in terms of prediction accuracy. SHAP value analysis further reveals that the exploration and exploitation of digital resources are the most influential predictors of idea implementation. The EEVC framework advances open innovation theory by clarifying how value cocreation dynamics influence idea implementation. Practically, it proposes a human–machine collaboration system that enhances expert decision-making for more effective idea selection.

1. Introduction

New product development (NPD) is a critical gateway process for open innovation initiatives. However, it often entails high costs and a meager success rate in identifying valuable ideas [1,2]. Collaborative crowdsourcing communities have emerged as widely used open innovation platforms, significantly reducing the cost of acquiring ideas [3]. However, they also increase the complexity of idea screening during the fuzzy front-end stage of NPD [4]. As a time-consuming and inherently uncertain task [5], ideal screening remains a major challenge for firms seeking to identify valuable ideas amid the vast amount of data. On the one hand, the dynamic and fast-evolving nature of product markets makes it difficult to predict which ideas will succeed in the long term [6]. On the other hand, the expanding pool of ideas, often coupled with rich networks of user-generated content and discussions, leads to information overload. This overload makes it even more challenging to identify promising ideas efficiently [7,8].

This information overload poses a critical problem for firms attempting to make optimal selection decisions [7,8,5]. The overwhelming volume of ideas and the complexity of evaluating their potential for successful implementation require a new approach. Therefore, the pressing research question is as follows: How can firms effectively manage the large volume of potentially valuable data to make decisions with greater accuracy and timeliness in idea selection? We hypothesize that an effective solution lies in a human–machine collaboration workflow [9]. In such a workflow, machine learning algorithms can filter large numbers of ideas and extract the most relevant ideas, which can then be further evaluated by human experts. This human–machine collaboration would allow firms to efficiently navigate the challenges of information overload, leveraging big data and machine learning to automate the initial screening process while preserving human expertise for final selection decisions [10,11].

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Previous studies have addressed idea screening from three main perspectives: content-based (e.g., [12,11]), contributor-based (e.g., [13,14]), and crowd-based (e.g., [15,16]) approaches. Hoornaert et al. [17] integrated these perspectives into a comprehensive model, the 3Cs model, to predict the likelihood of idea implementation. While the 3Cs model has advanced the field, it focuses primarily on static features of ideas and their contributors. This study goes beyond the 3Cs model by introducing Grönroos' [18] value cocreation model. This model emphasizes dynamic interactions among multiple participants (firm experts, users, and physical systems) and creative outcomes in the collaborative crowdsourcing process. Additionally, based on the fluidity and knowledge evolution mechanisms in collaborative crowdsourcing [19], we extend the value cocreation model by introducing community-accumulated digital resources as the creative context for value cocreation activities. Finally, this study incorporates March's [20] exploration and exploitation framework. In doing so, we propose that a value cocreation system that balances exploration and exploitation is more likely to produce ideas that are novel, feasible, and commercially valuable [21,22], thus increasing the likelihood of successful implementation.

Following the design science research paradigm [23] of information system design theory [24], we use Grönroos' [18] value cocreation model and March's [20] exploration and exploitation framework as the kernel theory to explain idea implementation. For meta-requirements, we develop the exploration and exploitation of value cocreation (EEVC) theoretical framework. This framework consists of six theoretical dimensions: the exploration or exploitation of digital resources, direct interactions, and ideas and their comments. With respect to meta-design, we construct a comprehensive set of computable dimensions and features for the EEVC. To test the efficacy of our model, we collected 110,384 idea threads from an open innovation community (Xiaomi's MIUI forum). We constructed two models for predicting idea implementation: the proposed EEVC-based idea-screening model and the well-regarded 3Cs model from the open innovation literature [4]. We compared the model performance on two design evaluation tasks and found that the EEVC-based model outperforms the 3Cs model, achieving state-of-the-art performance. To validate the robustness of our findings, we evaluated the models using a new MIUI dataset as well as a dataset from the Huawei Product Design Community. The results again show that the EEVC-based model consistently outperforms the 3Cs baseline model. Finally, we applied SHAP analysis to understand the advantages of the proposed model and the importance and specific effects of each feature. Additionally, we selected features to identify a quasi-optimal subset for idea screening.

This study makes several notable contributions to the open innovation literature. First, we propose, implement, and evaluate a theory-driven machine learning framework following the design science paradigm to provide machine recommendation support for the predictive task of idea screening for collaborative crowdsourcing of open innovation. We assess the superiority of the EEVC-based explainable machine learning model, which contributes to addressing the challenges of information overload and prediction difficulties in idea screening [11]. Second, we propose and emphasize the importance of digital resources as the creative context for value cocreation activities [25,26]. In doing so, we enrich the existing understanding of value cocreation systems and idea implementation. Third, our work contributes to the open innovation literature by first analyzing value cocreation from the perspective of exploration and exploitation [21,5]. We advance the conceptualization and operationalization of the EEVC in online collaborative innovation, laying a theoretical and empirical foundation for future research.

2. Theoretical background

2.1. Idea selection in collaborative crowdsourcing for open innovation

In the fuzzy front end of NPD, firms need to emphasize idea quality rather than acquiring a large number of ideas, as high-quality, original ideas are key to NPD success [27]. If the "wrong" idea is chosen at the outset, a firm's limited resources will be wasted on R&D projects destined to fail [28]. Throughout the NPD process, idea selection is the main link between idea generation and implementation [29]. Idea selection is a funnel-shaped process that allows a small number of good ideas to be selected while many poor ideas are eliminated. Essentially, idea selection is a predictive task [5]. In the face of great uncertainty about the future, companies need to make predictions from a pool of alternative ideas, i.e., "which ideas have the potential to contribute most to the success of NPD" [6]. Accurately predicting the value of ideas is inherently tricky and even more challenging if the set of alternative ideas is massive [7,5].

In particular, idea-screening activities in collaborative crowdsourcing communities are complex [30] because the parallel search strategy of online open innovation dramatically expands the set of ideas available for selection [31]. Furthermore, open innovation typically involves an extensive network of information from users who strongly argue about idea candidates [8]. Extreme information overload raises the much-discussed question of how firms can effectively process the vast amount of potentially valuable data to make the best selection decisions for NPD.

To cope with the pressure of information overload, communities introduce crowd evaluation systems that allow the crowd to participate in the idea-screening process. However, the central problem with crowd evaluations is that they tend to grossly underestimate the cost of idea execution while vastly overestimating the idea's potential [32]. In addition, crowd wisdom and corresponding idea quality are highly variable, resulting in much repetitive, low-quality, or misleading information [13,1]. This result not only misaligns the response of the crowd assessment to actual market demand but also further increases the pressure of information overload on firm experts to make selection decisions. Moreover, the crowd participation rate is typically extremely low, which triggers a systematic problem of "errors of omission" [33].

In essence, both selection decisions by firm experts and potential market tests by the crowd are manual assessments that are inevitably subject to bias and subjectivity [34]. Moreover, manual assessments are limited by limited information processing capacity, and individuals often suffer from information overload, which triggers cognitive overload [14]. One promising solution to this problem is to build an idea-screening system by using big data and machine learning technologies.

2.2. Related research on idea screening

Leveraging machine learning tools, previous research has established idea-screening models and systems from three key perspectives: content, contributors, and the crowd (see Table 1). **Content-based idea screening** uses machine learning algorithms to perform semantic analysis of idea content, facilitating the identification of high-quality ideas. Earlier studies have explored the concept of idea novelty through content similarity [35,12,36], while additional research has examined presentation characteristics [11] and the thematic content of ideas [37]. **Contributor-based idea screening** emphasizes the selection of optimal contributors, as high-quality ideas often stem from high-quality contributors [38,22]. Previous studies have aimed to identify these contributors by analyzing their past submissions, implementations, comments, and tenure within the community [13,14,25,39]. **Crowd-based idea screening** relies on votes and comments as critical metrics [16]. These metrics include diversity-based ranking [15] and the "lemon bag" voting method [40].

Hoornaert et al. [4] combined three idea-screening methods from previous studies to develop the **3Cs model** for predicting idea implementation (Table 1). The 3Cs model extracts features from content-based, contributor-based, and crowd-based screening methods. In addition, a new method was introduced to quantify the distinctiveness of ideas, and a nonlinear machine learning algorithm was applied to predict the likelihood of idea implementation. Their study revealed that

Table 1
Literature on idea screening.

		3Cs Idea-Screening Model [4]							
		Content		Contributor experience				Crowd feedback	
		LSI components	Relative distinctiveness	Number of past ideas	Number of previous contributor comments	Number of past implemented ideas	Tenure	Number of crowd votes	Number of crowd comments
Content-based idea screening	Novelty -Similarity [35,12,36]		√						
	Presentation [11]	X							
	Theme analysis [37]	√							
Contributor-based idea screening	Previous submissions [13,14]			√					
	Previous implementation [13,14,39]					√			
	Previous comments [13]				√				
	Tenure [14]						√		
Crowd-based idea screening	Votes -Diversity-based ranking system [15] -“Lemon bag” voting method [40]							√	
	Comments [16]								√

Note. √: included; X: not included.

considering content and contributor metrics improved ranking performance by 22.6 %–26.0 % and that adding crowd metrics improved ranking performance by 48.1 %. As shown in Table 1, the 3Cs model is a comprehensive predictive framework that integrates the three perspectives and associated predictive factors proposed in earlier research, excluding the presentation characteristics of ideas. Therefore, our study adopts the 3Cs model as the baseline model.

2.3. Value cocreation model with exploration and exploitation concepts

The **value cocreation model** developed by Grönroos [18] asserts that value cocreation involves joint actions between customers and service providers during direct interactions. From the perspective of the service provider, the value creation process flows from right to left (as illustrated in Fig. 1). The starting point is the customer’s participation in the service process, facilitating direct collaborative interactions. The resource categories involved include physical resources, contact employees, and customers (which can be either individual persons or groups of customers present simultaneously).

In Fig. 1, the area within the dotted circle represents the value cocreation platform. In the platform, direct interactions occur among the participating resource categories, including interactions between customers and physical systems and interactions between customers and contact employees. The former refer to obtaining useful customer

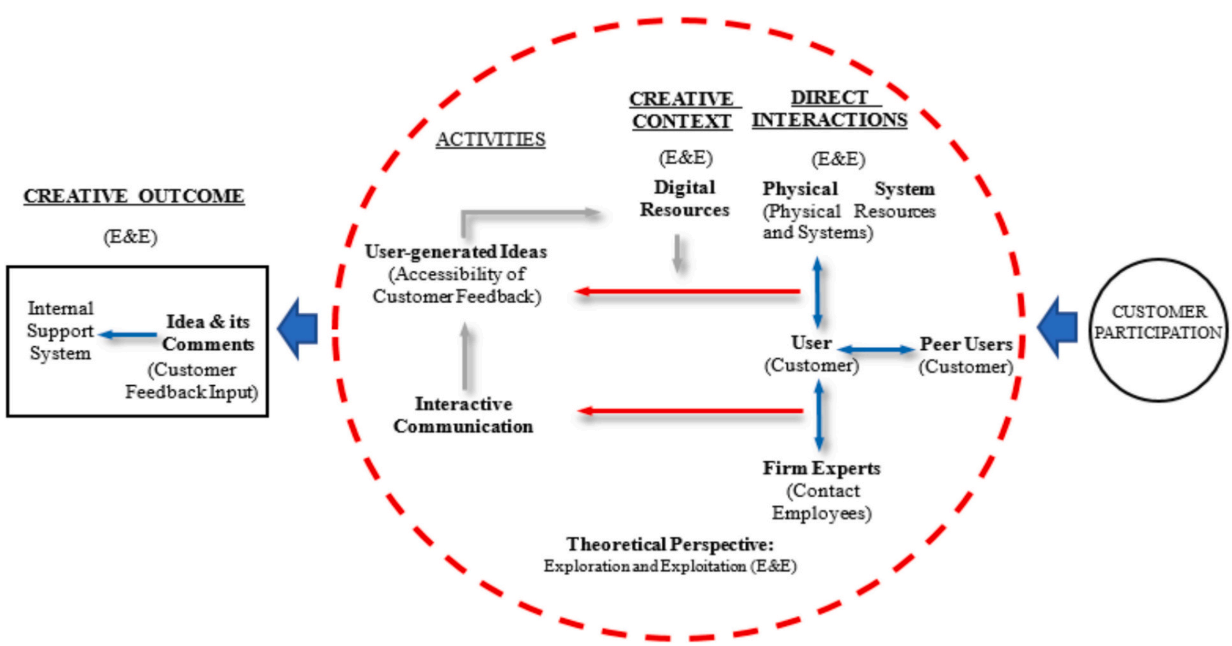

Fig. 1. An EEVC-based model for collaboration crowdsourcing for open innovation.
Note: E&E refers to exploration or exploitation; Source: Adapted from Grönroos [18].

feedback as customers utilize the physical systems provided by the company. The latter pertain to customer feedback generated during the communicative interactions between customers and contact employees. The entire value creation process involves the input of customer feedback. In addition, it requires an adequately prepared service provider to receive this feedback and convert it into actionable information through an internal support system, thus driving continual improvements in the service experience. Through these direct interactions, customers and service providers jointly create value relative to each party's needs.

The concepts of **exploration and exploitation** originate from the field of organizational learning. Exploration refers to the pursuit of new knowledge, technologies, and ideas, characterized by uncertainty in returns and a significant distance in time and space from the locus of action and adaptation [20]. This typically involves experimentation and innovation, which may lead to failures in the short term but are crucial for driving organizational innovation and differentiation in the long run [41]. In contrast, exploitation focuses on optimizing existing knowledge, technologies, and processes, viewed as a deepening use of current resources, with returns that are relatively stable and predictable [20]. The success of exploitation relies on enhancing existing capabilities and reducing uncertainty, which can lead to improved organizational performance in the short term [42]. However, an overemphasis on exploitation may result in the atrophy of innovative capabilities.

March [20] emphasizes that an appropriate balance between exploration and exploitation can promote short-term performance while supporting long-term innovation capabilities [43]. A rational resource allocation strategy can help organizations avoid the self-limiting constraints of a singular strategy, thus maintaining competitiveness in dynamic environments [44]. The existing empirical literature often measures these concepts as opposite ends of a continuum [45], which can obscure their distinct impacts and different configurations. Recently, scholars have advocated for separating exploration and exploitation to better understand their unique effects and the impacts of various combinations under different conditions [46]. This approach can provide clearer insights into their roles in organizational success and innovation.

3. Kernel theory-based design

To build a theory-driven machine learning system as an idea-screening model, we follow information system design theory to guide our system design process [23,24], which includes kernel theory, meta-requirements, and meta-design, as shown in Table 2.

3.1. Kernel theory

Based on Grönroos' [18] value cocreation model, we conceptualize collaborative crowdsourcing for open innovation as a value cocreation activity and outcome achieved through the joint participation of users and firms. In collaborative crowdsourcing communities, users contribute to the development of new products or services by continuously generating and submitting new ideas [47]. Firms adopt valuable user ideas and incorporate them into new products, ultimately creating offerings that better meet user needs or preferences [48]. This collaborative innovation enables users and corporate experts to communicate across geographic and temporal boundaries, engaging in joint value cocreation [49].

Furthermore, we introduce March's [20] exploration and exploitation framework as a theoretical perspective for analyzing the value cocreation model. To balance short-term economic returns with long-term competitive advantage [5,1], firms should holistically evaluate the novelty, feasibility, and commercial value of initial ideas during the fuzzy front-end stage of NPD and make implementation decisions accordingly [21,22]. Following the theoretical logic of exploration and exploitation, we propose that a value cocreation system that both explores new possibilities and leverages established certainties is more likely to generate ideas that meet comprehensive evaluation standards for implementation.

3.2. Meta-requirements

In Fig. 1, the area within the dotted lines represents the processes and activities of value cocreation within collaborative crowdsourcing communities. Unlike the well-defined customer service scenarios described by Grönroos [18], online collaborative innovation challenges involve temporary crowds, where participants continually join and leave the community [19]. This fluidity is a fundamental trait of online collaboration, enabling transient crowds to engage in value cocreation through mechanisms of knowledge accumulation [19]. Specifically, various contents generated during early-stage value creation activities gradually accumulate to form the community's digital resources, which, in turn, serve as the ***creative context*** for subsequent value cocreation activities [25,26]. Unlike fixed physical resources, the digital resources within the community are endogenous and cumulative, continuously evolving to establish a knowledge base for future value cocreation activities.

Based on Grönroos' [18] value cocreation process model (see Fig. 1), collaborative innovation begins with user participation in idea generation, allowing ***direct interactions*** between physical systems, users, and firm experts. User interactions with the physical system involve utilizing community-provided functions to generate and submit ideas or feedback. Interactions between users, their peers, and firm experts entail debates, evaluations, and the filtering of ideas or feedback. These interactive exchanges form a user-centered information network that encompasses both information input and output [50,51]. Ultimately, continuous value creation activities yield ***creative outcomes***, such as community-curated ideas and comments, which are subsequently recommended to an internal support system for corporate expert decisions on potential implementation in NPD.

Following the logic of exploration and exploitation, we establish six theoretical dimensions across three levels—the creative context, direct interaction, and creative outcomes—to predict the likelihood of idea implementation: the exploration or exploitation of digital resources, the

Table 2
Design Framework for the EEVC-based Idea-Screening Model.

Kernel theory	● Collaborative crowdsourcing in open innovation is conceptualized based on Grönroos' [18] value cocreation model. ● The impact of value cocreation processes and outcomes on idea implementation is analyzed from the perspective of March's [20] exploration and exploitation framework.
Meta-requirements	● Collaborative crowdsourcing for open innovation is operationalized as an extended value cocreation model and distinguishes three levels: the creative context constructed by community digital resources; direct interactions between physical systems, users, and firm experts; and the creative outcome composed of ideas and their comments. ● The EEVC has six theoretical dimensions: the exploration or exploitation of digital resources, the exploration or exploitation of direct interactions, and the exploration or exploitation of ideas and their comments.
Meta-design	● A comprehensive set of computable dimensions and features is constructed to represent the EEVC. ● An idea-screening system is designed that extracts features and predicts the likelihood of idea implementation.

Table 3
EEVC-based idea-screening model for the likelihood of idea implementation.

Value Cocreation	Exploration and Exploitation	Definition	Computable Dimensions
Creative context	D1–1: Exploration of digital resources	The exploration of digital resources refers to community-documented idea resources aimed at proposing new features or discovering new applications for existing features.	Exploration of implemented ideas Exploration of unimplemented ideas Exploration of ideas of the same class Exploration of popular ideas of the same class Exploration of ideas on the same day
	D1–2: Exploitation of digital resources	The exploitation of digital resources refers to community-documented idea resources focused on continuously optimizing current features and applications.	Exploitation of implemented ideas Exploitation of unimplemented ideas Exploitation of ideas of the same class Exploitation of popular ideas of the same class Exploitation of ideas on the same day
Direct Interactions	D2–1: Exploration of direct interactions	The exploration of direct interactions involves users engaging in new task classes or new forms of interaction with firm experts or peers.	Exploration of the user–physical system interactions Exploration of the user–firm expert interactions Exploration of the user–peer user interactions
	D2–2: Exploitation of direct interactions	The exploitation of direct interactions involves users deepening their engagement with familiar task classes or established interactions with firm experts or peers.	Exploitation of the user–physical system interactions Exploitation of the user–firm expert interactions Exploitation of the user–peer user interactions
Creative Outcomes	D3–1: Exploration of ideas and associated comments	The exploration of ideas and their comments involves developing new functions or new applications for existing features.	Exploration of ideas Exploration of ideas' comments
	D3–2: Exploitation of ideas and their comments	The exploitation of ideas and their comments focuses on optimizing and enhancing existing features and applications.	Exploitation of ideas Exploitation of ideas' comments

exploration or exploitation of direct interactions, and the exploration or exploitation of ideas and their comments (Table 3). We refer to this theoretical framework as the EEVC model, an extended value cocreation model that focuses on proposing new functionalities, finding new applications for existing functionalities, or optimizing current functionalities and applications.

3.3. *Meta-design*

3.3.1. *Exploration or exploitation of digital resources*

The exploration of digital resources refers to community-documented idea resources aimed at proposing new features or discovering new applications for existing features. In contrast, the exploitation of digital resources refers to community-documented idea resources focused on continuously optimizing current features and applications. This process reflects the community and corporate preferences for innovation. The preferences for exploration or exploitation signaled by the community and corporation trigger observational learning by users, either consciously or unconsciously [52], influencing their subsequent idea generation.

The assessment and categorization of digital resources provide precise signals of community and corporate preferences. Specifically, implemented and unimplemented ideas serve as examples of success or failure [13,25], indicating the preferences of firm experts. Ideas within the same class focus on consistent task characteristics [14], reflecting specific task category preferences. High-popularity ideas are widely followed [39], indicating the preferences of user groups. Same-day ideas are generated in the same period [50], showing the preferences of active creators. Thus, we propose that the exploration and exploitation of these various digital resources impact the likelihood of idea implementation.

3.3.2. *Exploration or exploitation of direct interactions*

The exploration of user–physical system interactions involves users trying new task classes. In contrast, the exploitation of these interactions involves users delving deeper into previously engaged task classes. Exploring new tasks broadens the knowledge domain and increases opportunities for users to reorganize knowledge in novel ways, enhancing idea novelty [53]. Exploiting old tasks promotes knowledge reuse within established domains, helping users identify risks and filter flawed solutions and thus improving idea feasibility [54]. Therefore, we propose that the exploration or exploitation of user–physical system interactions influences the likelihood of idea implementation.

The exploration of user–firm expert and user–peer interactions refers to users engaging with information networks to explore new ideas. The exploitation of these interactions involves applying and analyzing existing ideas within these networks. Perry-Smith [55] explains that knowledge communication can enhance creativity from both the transfer and constructivist perspectives. From the transfer perspective, users apply the exploration and exploitation materials of information networks to generate ideas [56]. From the constructivist perspective, exploring and exploiting information networks reshape users' interpretation, formulation, and framing of creative tasks, influencing their idea-generation processes [57]. Therefore, we propose that the exploration or exploitation of user–firm expert and user–peer user interactions influences the likelihood of idea implementation.

3.3.3. Exploration or exploitation of ideas and their comments

The exploration of ideas and their comments pertains to the innovative qualities of creative outcomes, aiming to propose new features or discover new applications for existing features. The exploitation of ideas and their comments focuses on continually optimizing current features and applications. Exploring ideas and comments involves developing new functions or applications with a high degree of novelty. In contrast, exploiting ideas and comments focuses on optimizing existing features or applications and enhancing compatibility and feasibility. Chung et al. [58] examined how the nature of innovation (explorative versus exploitative) in a firm's patent portfolio affects firm value. Hence, we propose that the exploration or exploitation of ideas and their comments influences idea implementation.

As shown in Table 3, we capture a rich set of features based on the EEVC for predicting the likelihood of idea implementation. We also compare the dimensions of the 3Cs model with those of the EEVC-based model in Table 4. The EEVC-based model encompasses more theoretical dimensions than the 3Cs model does, which only partially addresses the three computable dimensions of the EEVC-based model. Therefore, we speculate that the EEVC-based idea-screening system should outperform the 3Cs model.

4. Design instantiation: an idea-screening model

We present the design process of our proposed system aimed at predicting the likelihood of idea implementation. The framework, illustrated in Fig. 2, consists of three main components: (1) data collection, (2) feature extraction, and (3) model construction.

4.1. Step 1: data collection

We developed a web scraper to gather necessary information, such as idea threads, from an online open innovation community. Using the Python library BeautifulSoup, we parsed the crawled HTML documents to extract idea content, user information, responses from firm experts, comments from other users, and other relevant data. This information was stored in a relational database, excluding HTML tags and metadata. Firm experts' responses (e.g., whether to adopt an idea) represent the outcome for each idea, while other characteristics serve as predictors. Finally, data tuples (idea-related information and implementation outcomes) were used to train the idea-screening model.

4.2. Step 2: feature extraction

To train our proposed model, we extracted features from idea-related information (e.g., idea content, idea comments, ideator past behaviors, and firm expert responses). Specifically, we applied semantic analysis, linguistic inquiry, text similarity, and statistical techniques to generate our proposed features.

For **semantic analysis**, we used various related methods, including the LDA model [59], doc2vec model [60], and BERT model [61], to extract various semantic dimensions of textual content. For example, LDA is used to discover the main topic components related to customer requirements in each idea, which is similar to but better than the LSI components in the 3Cs model [4]. To determine the topic number k, we follow Griffiths and Steyvers [62] and calculate the coherence scores of LDA models with different topic numbers $k \in \mathrm{K} = \{5, 6, 7, \ldots, 25\}$. We finally set $\mathrm{K} = 14$ because the LDA model obtains the highest coherence score with this setting. More details, including the top ten representative words for each topic, are shown in Fig. I and Table I (see the appendix). In addition, we apply the doc2vec and BERT language models, which are based on deep learning, to convert a sentence or paragraph to a distributed semantic vector. These idea embeddings, which are the semantic representations of ideas, can be used to calculate textual similarity-related variables.

In this study, many variables, especially the "similarity-based" variables in Table V (see the appendix), are measured based on **text similarity**. In particular, we adopt cosine similarity to quantify the textual similarity between ideas. The formula of cosine similarity is as follows, in which A and B are the text representations of ideas:

$$\text{similarity}(A, B) = \frac{A \bullet B}{\|A\| \times \|B\|}.$$

To calculate the text similarity between ideas, we first segmented Chinese sentences into phrases using the JieBa segmentation package. Second, we tokenized all ideas i (implemented or unimplemented ideas) into a vector space representation. Third, we applied the doc2vec and BERT[1] models to estimate idea embeddings. Finally, according to different definitions, we measured the similarity-based variables, such as "similarity within ideas of the same class", "similarity within popular ideas of the same class", "similarity to implemented ideas", "similarity within implemented ideas", and "relative distinctiveness (in the 3Cs model)".

Another important method for feature extraction is **linguistic inquiry**. Specifically, we adopted Linguistic Inquiry and Word Count (LIWC) software to generate the "lexicon-based" variables in Table V (see the appendix). LIWC is text-analysis software featuring a group of built-in dictionaries based on the Java programming language [63]. Following the work of Uotila et al. [64] and March [20], we constructed two word lists for exploration and exploitation. For example, exploration consists of "search", "variation", "risk taking", "experimentation", "play", "flexibility", "discovery", and "innovation"; exploitation includes "refinement", "choice", "production", "efficiency", "selection", "implementation", and "execution". Finally, based on the two word lists and LIWC, we calculated lexicon-based variables, such as "exploration of implemented ideas", "exploitation of implemented ideas", "exploitation of unimplemented ideas", and "exploration of ideas of the same class".

Finally, we used simple statistical techniques to measure the remaining variables, including "degree of involvement in idea classes", "contributor experience", and "crowd feedback". After all the features were extracted, we converted each idea into a tuple (x_i, y_i), where x_i is a feature vector extracted from idea-related information and y_i is the outcome of each idea.

4.3. Step 3: model construction

Third, we trained and tested the idea-screening model. Specifically, a tuple set $\{(x_i, y_i), i \in N$, where N is the set of all ideators$\}$ was randomly selected and split into a training set and a testing set. Next, we adopted six classic supervised learning algorithms to train the proposed model: the AdaBoost, logistic regression, support vector machine (SVM), random forest, convolutional neural network (CNN), and XGBoost algorithms.[2] Based on the training dataset, each learning algorithm was used to fit the objective function and generate an idea-screening model. Finally, the trained model was assessed and evaluated via the test dataset. More details are discussed in Section 5.

[1] We used two approaches to generate the idea-text representation, and the model with the BERT representation outperformed that with the doc2vec representation. Thus, the final results in this paper are based on the BERT representation.

[2] We applied the Python package Keras to implement a CNN and scikit-learn to implement the AdaBoost, logistic regression, SVM, random forest, and XGBoost algorithms.

Table 4
EEVC-based idea-screening model vs. 3Cs idea-screening model.

	Baseline: 3Cs model	EEVC-based Idea-Screening Model		
		D1: Exploration or exploitation of digital resources	D2: Exploration or exploitation of direct interactions	D3: Exploration or exploitation of ideas and their comments
3Cs Idea-Screening Model				
Content	√	X	X	※
Contributor experience	√			
Crowd feedback	√			
EEVC-based Idea-Screening Model				
Digital Resources	√	√		
Direct Interactions	√		√	
Ideas and their Comments	√			√

Note. √: included; X: not included; ※: partially included.

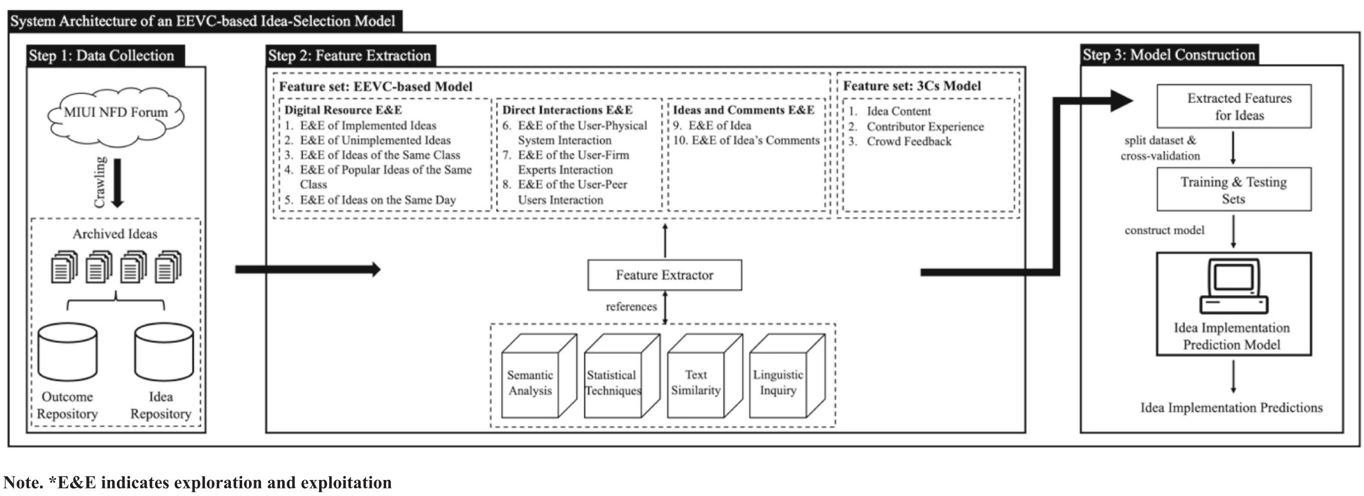

Fig. 2. System architecture of the EEVC idea-screening model.
Note. *E&E indicates exploration and exploitation.

5. Design evaluation setting

5.1. Evaluation data

To validate the proposed model, we collected data from Xiaomi's MIUI forum, a prominent firm-hosted open innovation community in China. Established by Beijing Xiaomi Technology, Inc., Xiaomi became the world's third-largest smartphone manufacturer by the third quarter of 2020, after Samsung and Huawei. The MIUI forum invites users to submit ideas for optimizing or adding functions to the MIUI mobile phone operating system. The effective integration of user ideas with Xiaomi's resources fosters continuous innovation, creating products that meet diverse user needs and align with user habits.

We collected all idea threads from the MIUI forum from August 2010 to December 2015, totaling 110,384 ideas. We preprocessed the data by deleting special posts, including 515 posts from the MIUI Official Team, moderators, or the MIUI Designer Team; 578 posts without a submission time; 217 summary posts; and 3 posts with missing information. Additionally, we removed 19,165 posts without idea descriptions. After preprocessing, 89,906 ideas remained. Summary statistics for the data are shown in Table II (see the appendix).

5.1.1. Measurement of idea implementation

Training supervised learning algorithms requires data labeled with idea implementation outcomes (e.g., implemented or not). These labels indicate the results of comprehensive measurements of an idea's innovation value. In the MIUI forum, internal experts (members of the MIUI development group) provide feedback by affixing a seal to the upper-right area of the thread (see Table III in the appendix). We assigned a value of 1 to implemented ideas and 0 to unimplemented ideas. Table IV (see the appendix) summarizes the annual statistics of idea outcomes. In total, 6218 ideas (6.9 %) were implemented, whereas 83,688 ideas (93.1 %) were not.

5.1.2. Measurement of features in the EEVC-based idea-screening model

Table V (see the appendix) summarizes the measurement of features in the EEVC-based idea-screening model. We measure these features using methods such as textual similarity, linguistic inquiry and word count, LDA, the linguistic category model, and idea categories (see Table VI in the appendix).

5.2. Evaluation setup

As shown in Fig. 3, we designed two tasks for predicting idea implementation. In the first task, we used all ideas posted before 2015 to train and test the idea-screening models. The data are highly imbalanced, with 4366 implemented ideas versus 53,211 unimplemented ideas, as most ideas are not implemented by firm experts. Japkowicz and Stephen [65] claim that high data skewness in a training set can lead to poor prediction or classification performance. To mitigate the negative

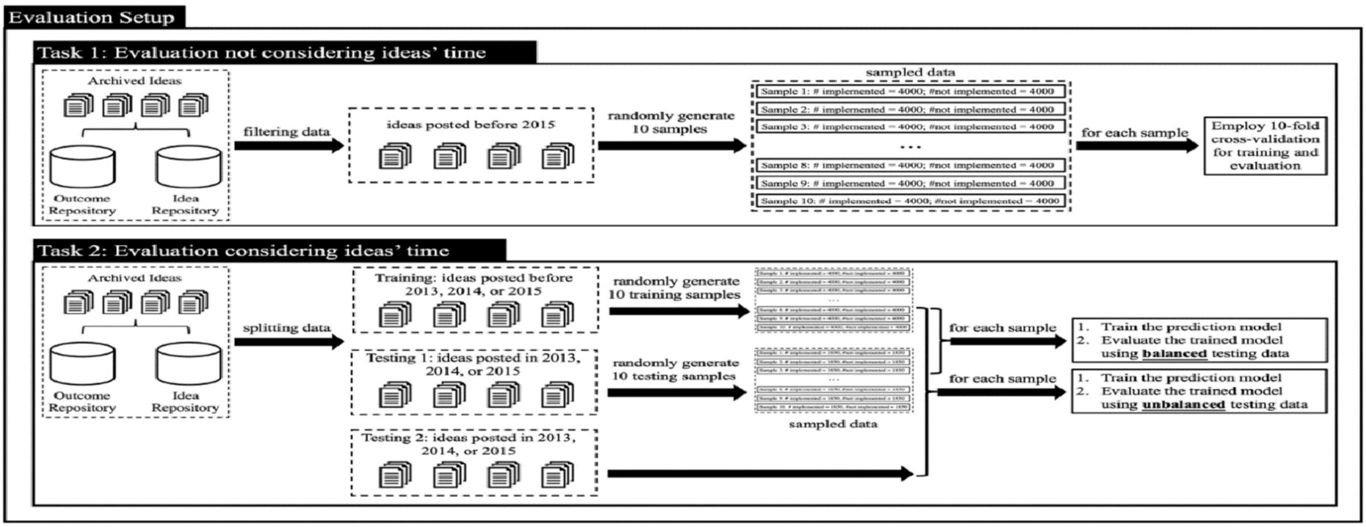

Fig. 3. Evaluation setup.

Table 5
Three scenarios of data splitting.

	Dates of Data Splitting	Ideas in the Preperiod Data for Training	Ideas in the Postperiod Data for Testing
Scenario 1	Jan. 1st, 2013	ideas posted before 2013 28,278 ideas (1625 implemented and 26,653 unimplemented)	ideas posted in 2013 11,358 ideas (833 implemented and 10,525 unimplemented)
Scenario 2	Jan. 1st, 2014	ideas posted before 2014 39,636 ideas (2458 implemented and 37,178 unimplemented)	ideas posted in 2014 18,078 ideas (1908 implemented and 16,170 unimplemented)
Scenario 3	Jan. 1st, 2015	ideas posted before 2015 57,577 ideas (4366 implemented and 53,211 unimplemented)	ideas posted in 2015 32,329 ideas (1852 implemented and 30,477 unimplemented)

effect of data imbalance, we applied an undersampling strategy[3] [66,17] to reduce instances of the majority class. Specifically, we randomly selected ten samples, each containing 4000 implemented (positive) and 4000 unimplemented (negative) ideas. For each sample, we used tenfold cross-validation to train and test the baseline and proposed models. However, this approach does not account for the time when ideas are posted. In reality, we cannot use later ideas (in the training set) to predict earlier ideas (in the testing set). Additionally, the performance of supervised learning methods might be overestimated, as later ideas (in the training set) can provide extra information for idea screening.

To address the hindsight issue in Task 1, Task 2, in which we used all the data, was developed to strictly divide the training and testing sets using three specific dates: Jan. 1st, 2013; Jan. 1st, 2014; and Jan. 1st, 2015. Table 5 shows more details about the three strategies of data splitting. For the three data-split dates, there are 28,278 ideas (1625 implemented and 26,653 unimplemented), 39,636 ideas (2458 implemented and 37,178 unimplemented), and 57,577 ideas (4366 implemented and 53,211 unimplemented) in the preperiod data. Additionally, there are 11,358 ideas (833 implemented and 10,525 unimplemented), 18,078 ideas (1908 implemented and 16,170 unimplemented), and 32,329 ideas (1852 implemented and 30,477 unimplemented) in the postperiod data. Clearly, the dataset is not balanced. Thus, we again applied undersampling to randomly generate ten data samples for the training set, where each sample consisted of identical numbers (1600, 2400, and 4000) of implemented and unimplemented instances. Unlike Task 1, we generated two types of testing sets: balanced and unbalanced. In the balanced set, we randomly selected ten testing samples, each containing the same number of implemented and unimplemented instances (833, 1908, and 1852). In the unbalanced set, we maintained the original data distribution to reflect the real-world scenario [67].

Based on the setting, we experimented to confirm the efficacy and quality of the proposed model. To mathematically evaluate the performance of the baseline model and the proposed model, we adopted seven commonly used metrics for information retrieval: accuracy, precision, recall, the F1 score, the area under the receiver operating characteristic curve (AUC), the Spearman rank-order correlation (SRC), and the normalized discounted cumulative gain (NDCG). Following Liu et al. [37], all the method performance results are presented in the evaluation results section in Table VII (see the appendix).

6. Design evaluation results

6.1. EEVC-based model versus the baseline

To demonstrate the advantages of our proposed model, we used Python to construct the baseline and proposed models with some variations. First, we followed Hoornaert et al. [4] by building the 3Cs model as the baseline model. We also built incremental models (e.g., baseline + D1, baseline + D2, and baseline + D3) by adding each part of our EEVC-based idea-screening model to identify the contribution of each part of the EEVC to idea screening. Finally, we ran the full proposed model (baseline + D1 + D2 + D3).

Following the evaluation setting, the performance results of Task 1 are given in Table 4, in which precision, recall, the F1 score, accuracy, the AUC, and the SRC are reported (NDCG is reported in Appendix Table VIII). The paired *t*-test results demonstrate that the proposed model outperformed the baseline model in all supervised learning algorithms and that each independent part based on the EEVC is critical for improving idea screening. In particular, XGBoost performed best in all seven metrics. Additionally, the proposed model using XGBoost

[3] There are other strategies to reduce the impact of data skewness, such as oversampling, synthetic sampling, and cost sensitivity learning. However, sampling strategies are not our main focus; thus, we do not investigate them in this study. Oversampling was also applied to our main analysis; similar results were obtained. To shorten the length of the paper, only the results based on undersampling are presented.

significantly outperformed the baseline model in all seven metrics by 12.4 %, 18.5 %, 15.2 %, 14.3 %, 15.8 %, 53.4 %, and 3 %. To further validate the superiority of the 3Cs model based on the EEVC framework for idea screening, we incorporated three additional baseline models recently published by Qin et al. [68], Dahlander et al. [69], and Hsiang and Rayz [70]. As shown in Appendix Tables IX, X, and XI, the 3Cs baseline model outperformed the three newly added baseline models, and the proposed EEVC-based model significantly outperformed the three additional models and the 3Cs baseline.

As mentioned previously, hindsight exists in the setting of Task 1. To reduce the impact of the hindsight issue in Task 1, we used the evaluation setting in Fig. 3 to execute Task 2. Based on the performance in Task 1, we applied only XGBoost to the baseline and our proposed models in Task 2 to simplify the presentation of results. As shown in Fig. 3 and discussed in Subsection 5.2, we have three different ways to generate the training and testing datasets: Scenario 1, Scenario 2, and Scenario 3. Accordingly, we ran the baseline method (3Cs model), baseline + D1, baseline + D2, baseline + D3, and the full proposed model (baseline + D1 + D2 + D3) on three different datasets. The results, presented in Table 5 (Scenario 3), again show that the proposed model performed significantly better than the baseline model did (Scenario 1 and Scenario 2 are shown in Table XII of the appendix). In the case of the balanced testing data, the proposed model easily outperformed the baseline model with no surprises on the three different datasets, although its absolute performance decreased slightly relative to that on Task 1. The proportional increase was even larger than our model's performance in Task 1. Specifically, in Scenario 3, which had the best performance, our full model outperformed the baseline model on all seven metrics (precision, recall, F1 score, accuracy, AUC, SRC, and NDCG) by 23.4 %, 22.3 %, 22.2 %, 22.3 %, 28.5 %, 182 %, and 3.5 %, respectively. In addition, each part of the EEVC provides a critical contribution to idea screening because the baseline + D1, baseline + D2, and baseline + D3 approaches outperformed the baseline method (3Cs model) in most cases. Finally, D3 and D1 make the greatest contributions to the idea-screening model.

For the unbalanced testing dataset, the advantage of the proposed model was sustained. In Scenario 3, our model outperformed the baseline model on all seven metrics (precision, recall, F1 score, accuracy, AUC, SRC, and NDCG) by 5.6 %, 22.3 %, 17.6 %, 36 %, 28 %, 115 %, and 7.9 %, respectively. Furthermore, the D1, D2, and D3 components contributed significantly to predicting idea implementation in most cases. Surprisingly, the deep learning method, the CNN, failed to deliver the best performance in both Tasks 1 and 2. The poor performance of the CNN might be caused by the small training set, which could lead to overfitting in the training process. To further confirm the validity of our model, we ran another deep learning method, namely, long short-term memory (LSTM) [71], which uses the idea's pure content (the details of LSTM hyperparameter tuning are shown in Appendix Table XIII). As shown in Table 8, our proposed model performs much better than LSTM with the idea's textual content. The baseline (3Cs model) can also outperform LSTM with only idea text. Therefore, idea content alone is insufficient to predict idea implementation. The results confirm that our proposed features can provide more information than the idea content can. In summary, the results in Tables 6, 7, and 8 demonstrate the validity and effectiveness of EEVC model, which is particularly prominent in the significant contribution made by each component of the EEVC framework to idea screening.

6.2. Interpretability of the EEVC-based idea-screening model

Prior research highlights that good interpretability of design science research is crucial, as it helps identify comprehensive and meaningful features of the proposed model (theoretical explanations) and shows how each feature influences the model's estimation for a single prediction (technical explanations) [72]. To better understand the efficacy of the EEVC-based idea-screening model and demystify machine learning methods, we adopted the SHAP module to analyze and illustrate our model [35]. The SHAP module aims to explain the prediction of an instance x by computing the contribution of each feature. This tool calculates Shapley values, which are defined as the expected marginal contribution over all possible coalitions [2]. The module generates figures that provide theoretical and technical explanations for a given prediction or classification model.

As discussed in Section 3, the proposed model is driven by the EEVC framework. Based on this framework, we suggest 29 variables derived from three critical theoretical dimensions and apply computing techniques to quantify these variables. However, owing to the limited transparency of machine learning methods, we cannot directly determine the importance of the proposed features. Fortunately, SHAP mitigates this issue. Fig. 4 lists the top 20 features in terms of SHAP values, showing the most important features for idea screening. Only three of these features are from the baseline 3Cs model. The other 17 features, introduced by our study, are from the EEVC perspective, including 13 features from D1, 1 feature from D2, and 3 features from D3. Thus, our EEVC-related features significantly impact the prediction output, with the exploration and exploitation of digital resources (D1) having the most substantial predictive power.

This result indicates that the EEVC plays an essential role in idea screening, with the exploration and exploitation of digital resources being particularly critical as a creative context for value cocreation. The SHAP summary plots on the right side of Fig. 4 display the complex impact of each digital resource-related feature on predicting an idea's likelihood of implementation.

SHAP also provides technical explanations for specific predictions. Specifically, we used the shap.force_plot SHAP function to analyze a particular observation. Force plots illustrate how features contribute to the model's prediction for a specific observation, effectively explaining how a model arrived at its prediction. For example, in Fig. 5, we used the baseline model (3Cs model) and our proposed model (EEVC-based model) to predict whether a given observation would be "implemented" or "unimplemented", with the actual outcome being "implemented". This example demonstrates how our model enhances idea-screening performance and suggests that our proposed model provides additional predictive power. Several similar examples exist in our data, and the results in Tables 4 and 5 support this conclusion.

6.3. Feature screening and a quasi-optimal subset of features

Chandrashekar and Sahin [73] noted that since feature selection is an NP-hard problem, finding a globally optimal solution is impossible, requiring an almost infinite amount of time when the feature set is large. Therefore, our study aims to find a quasi-optimal feature subset that can further improve idea-screening performance [74,55] rather than solving the NP-hard problem. We adopted a filter-type feature selection algorithm to rank all features according to their SHAP values. The features were ranked in descending order of the SHAP value, and every five features were sequentially added to the model in steps. Finally, we obtained a quasi-optimal feature subset (the top 35 features by SHAP values). Table 9 shows the performance of the proposed model when different feature sets are used. The proposed model performs better with the quasi-optimal feature set than with the full feature set.

7. Discussion

Idea screening is a challenging and critical predictive task that significantly impacts the success or failure of NPD. Guided by the design

Table 6
Evaluation results of task 1.

Feature Sets	CNN			AdaBoost			Logistic Regression			SVM			Random Forest			XGBoost		
	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1
Baseline	0.618	0.649	0.632	0.63	0.684	0.653	0.621	0.667	0.643	0.621	0.685	0.651	0.625	0.662	0.643	0.630	0.670	0.650
	(0.019)	(0.032)	(0.017)	(0.017)	(0.018)	(0.013)	(0.017)	(0.019)	(0.014)	(0.017)	(0.020)	(0.014)	(0.018)	(0.017)	(0.014)	(0.017)	(0.019)	(0.015)
Baseline + D3	0.621	0.658	0.639	0.649*	0.709**	0.678**	0.655***	0.690**	0.672***	0.653***	0.744***	0.695***	0.649**	0.753***	0.696***	0.663***	0.748***	0.702***
	(0.021)	(0.091)	(0.042)	(0.014)	(0.015)	(0.011)	(0.019)	(0.022)	(0.018)	(0.016)	(0.019)	(0.014)	(0.022)	(0.023)	(0.014)	(0.015)	(0.018)	(0.012)
Baseline + D2	0.619	0.654	0.636	0.633	0.684	0.657	0.635	0.668	0.651*	0.637*	0.702*	0.667*	0.632	0.728***	0.676***	0.645*	0.724***	0.682***
	(0.020)	(0.082)	(0.033)	(0.024)	(0.022)	(0.017)	(0.022)	(0.020)	(0.015)	(0.018)	(0.025)	(0.013)	(0.018)	(0.025)	(0.016)	(0.018)	(0.017)	(0.014)
Baseline + D1	0.626*	0.701***	0.661***	0.642*	0.703**	0.671**	0.650**	0.708***	0.677***	0.658***	0.759***	0.705***	0.659***	0.776***	0.712***	0.668**	0.751***	0.707***
	(0.022)	(0.068)	(0.024)	(0.020)	(0.016)	(0.014)	(0.021)	(0.024)	(0.018)	(0.020)	(0.018)	(0.013)	(0.019)	(0.018)	(0.013)	(0.013)	(0.019)	(0.007)
Baseline + D1 + D2 + D3	0.634**	0.716***	0.673***	0.679***	0.728***	0.703***	0.674***	0.727***	0.699***	0.676***	0.782**	0.725***	0.685***	0.790***	0.734***	0.708***	0.794***	0.749***
	(0.023)	(0.037)	(0.021)	(0.016)	(0.015)	(0.009)	(0.018)	(0.019)	(0.014)	(0.019)	(0.018)	(0.012)	(0.016)	(0.013)	(0.010)	(0.013)	(0.017)	(0.006)
	Acc.	AUC	SRC	Acc.	AUC	SRC	Acc.	AUC	SRC	Acc.	AUC	SRC	Acc.	AUC	SRC	Acc.	AUC	SRC
Baseline	0.623	0.574	0.127	0.638	0.691	0.330	0.631	0.685	0.321	0.633	0.687	0.324	0.633	0.691	0.330	0.638	0.698	0.343
	(0.017)	(0.037)	(0.065)	(0.013)	(0.014)	(0.025)	(0.013)	(0.015)	(0.025)	(0.012)	(0.015)	(0.026)	(0.014)	(0.015)	(0.026)	(0.014)	(0.015)	(0.026)
Baseline + D3	0.625	0.637***	0.237***	0.661***	0.726***	0.390***	0.661***	0.715***	0.372***	0.672**	0.731***	0.400***	0.670***	0.734***	0.406***	0.681***	0.749***	0.431***
	(0.010)	(0.011)	(0.018)	(0.009)	(0.011)	(0.018)	(0.015)	(0.013)	(0.022)	(0.010)	(0.009)	(0.015)	(0.014)	(0.014)	(0.024)	(0.010)	(0.010)	(0.017)
Baseline + D2	0.592	0.638***	0.238***	0.641	0.700	0.347*	0.639	0.695	0.338	0.648	0.704	0.354	0.649*	0.711*	0.365*	0.657**	0.722**	0.384**
	(0.009)	(0.009)	(0.016)	(0.015)	(0.017)	(0.029)	(0.014)	(0.013)	(0.023)	(0.011)	(0.008)	(0.013)	(0.012)	(0.013)	(0.023)	(0.012)	(0.013)	(0.022)
Baseline + D1	0.627	0.674***	0.300***	0.654*	0.722***	0.383***	0.663***	0.722***	0.385***	0.687***	0.744***	0.422***	0.696***	0.756***	0.444***	0.671***	0.741***	0.418***
	(0.016)	(0.017)	(0.030)	(0.012)	(0.015)	(0.027)	(0.016)	(0.014)	(0.024)	(0.013)	(0.012)	(0.020)	(0.010)	(0.010)	(0.017)	(0.006)	(0.004)	(0.007)
Baseline + D1 + D2 + D3	0.655*	0.686***	0.317***	0.691***	0.761***	0.452***	0.685***	0.753***	0.428***	0.703***	0.762***	0.449***	0.717***	0.791***	0.509***	0.729***	0.808***	0.526***
	(0.014)	(0.019)	(0.024)	(0.007)	(0.011)	(0.019)	(0.010)	(0.009)	(0.016)	(0.010)	(0.008)	(0.014)	(0.008)	(0.011)	(0.015)	(0.006)	(0.007)	(0.013)

Due to space limitations, we present the NDCG results in Appendix Table VIII.
Note 1. Pre. = precision; Rec. = recall; F1=F1 score; Acc. = accuracy.
Note 2. Significant results compared with the baseline model are highlighted in bold, where * indicates $p < 0.1$; ** indicates $p < 0.05$; and *** indicates $p < 0.01$.
Note 3. The number in parentheses is the standard deviation of model performance in terms of the given metric.
Note 4. D1 is the exploration and exploitation of digital resources; D2 is the exploration and exploitation of direct interactions; and D3 is the exploration and exploitation of ideas and their comments.

Table 7
Evaluation results of task 2 (Scenario 3).

	Scenario 3: Data-Split date: Jan. 1st, 2015									
	Balanced Testing Dataset (10 samples) (Implemented ideas = 1852) (Unimplemented ideas = 1852)					Unbalanced Testing Dataset (Implemented ideas = 1852) (Unimplemented ideas = 30,477)				
	Baseline	Baseline + D3	Baseline + D2	Baseline + D1	Baseline + D1 + D2 + D3	Baseline	Baseline + D3	Baseline + D2	Baseline + D1	Baseline + D1 + D2 + D3
Acc.	0.564	0.611***	0.585***	0.624***	**0.690*****	0.508	0.609***	0.570***	0.624***	**0.691*****
	(0.008)	(0.008)	(0.007)	(0.009)	**(0.009)**	(0.029)	(0.006)	(0.007)	(0.008)	**(0.006)**
Pre.	0.565	0.614***	0.583*	0.625***	**0.698*****	0.514	0.524***	0.527***	0.533***	**0.543*****
	(0.008)	(0.008)	(0.007)	(0.009)	**(0.006)**	(0.001)	(0.001)	(0.001)	(0.002)	**(0.003)**
Rec.	0.564	0.611***	0.587***	0.624***	**0.690*****	0.565	0.609***	0.584**	0.624***	**0.691*****
	(0.008)	(0.008)	(0.006)	(0.009)	**(0.009)**	(0.007)	(0.006)	(0.006)	(0.007)	**(0.006)**
F1	0.562	0.609***	0.579***	0.623***	**0.687*****	0.393	0.419***	0.407*	0.437***	**0.462*****
	(0.009)	(0.009)	(0.009)	(0.010)	**(0.011)**	(0.014)	(0.010)	(0.007)	(0.011)	**(0.022)**
AUC	0.593	0.656***	0.611***	0.672***	**0.762*****	0.595	0.652***	0.637**	0.672***	**0.762*****
	(0.012)	(0.011)	(0.012)	(0.011)	**(0.009)**	(0.010)	(0.010)	(0.010)	(0.009)	**(0.007)**
SRC	0.161	0.269***	0.185**	0.298***	**0.454*****	0.098	0.128***	0.114**	0.148***	**0.211*****
	(0.020)	(0.019)	(0.021)	(0.020)	**(0.016)**	(0.007)	(0.006)	(0.007)	(0.006)	**(0.005)**
NDCG	0.923	0.941***	0.934**	0.939***	**0.955*****	0.711	0.729***	0.721*	0.732***	**0.767*****
	(0.004)	(0.004)	(0.004)	(0.004)	**(0.004)**	(0.004)	(0.004)	(0.004)	(0.005)	**(0.007)**

To show the robustness of our proposed model, we trained the model using old data and collected new data after May 2020 in the MIUI forum. The results show that our model can outperform the baseline methods. More details are included in Appendix Table XIV. In addition, we conducted a robustness test using data from the Huawei Product Design Forum. The results similarly confirm that our model outperforms the baseline approaches. Further details are presented in Appendix Table XV.
Note 1. Pre. = precision; Rec. = recall; F1 = macro F1 score; Acc. = accuracy.
Note 2. Significant results compared with the baseline model are highlighted in bold, where * indicates p < 0.1; ** indicates $p < 0.05$; and *** indicates $p < 0.01$.
Note 3. The number in parentheses is the standard deviation of model performance in terms of the given metric.
Note 4. D1 is the exploration and exploitation of digital resources; D2 is the exploration and exploitation of direct interactions; and D3 is the exploration and exploitation of ideas and their comments.

Table 8
Comparison between LSTM and Our Proposed Model in Scenario 3.

	Balanced Testing Dataset (10 samples) (Implemented ideas = 1852) (Unimplemented ideas = 1852)			Unbalanced Testing Dataset (Implemented ideas = 1852) (Unimplemented ideas = 30,477)		
	LSTM	Baseline	EEVC-based Idea-Screening Model	LSTM	Baseline	EEVC-based Idea-Screening Model
Acc.	0.512	0.564	**0.690*****	0.515	0.508	**0.691*****
	(0.005)	(0.008)	**(0.009)**	(0.003)	(0.029)	**(0.006)**
Pre.	0.469	0.565	**0.698*****	0.504	0.514	**0.543*****
	(0.143)	(0.008)	**(0.006)**	(0.009)	(0.001)	**(0.003)**
Rec.	0.362	0.564	**0.690*****	0.509	0.565	**0.691*****
	(0.191)	(0.008)	**(0.009)**	(0.003)	(0.007)	**(0.006)**
F1	0.394	0.562	**0.687*****	0.436	0.393	**0.462*****
	(0.142)	(0.009)	**(0.011)**	(0.077)	(0.014)	**(0.022)**
AUC	0.513	0.593	**0.762*****	0.514	0.595	**0.762*****
	(0.005)	(0.012)	**(0.009)**	(0.005)	(0.010)	**(0.007)**
SRC	0.175	0.161	**0.454*****	0.023	0.098	**0.211*****
	(0.222)	(0.020)	**(0.016)**	(0.005)	(0.007)	**(0.005)**
NDCG	0.908	0.923	**0.955*****	0.685	0.711	**0.767*****
	(0.005)	(0.004)	**(0.004)**	(0.003)	(0.004)	**(0.007)**

science paradigm, we developed the EEVC theoretical framework, based on which we built an innovative, interpretable machine learning-based idea-screening system. Conceptually, the EEVC framework posits that a value cocreation model that integrates both exploration and exploitation is more likely to generate ideas that meet comprehensive evaluation standards. We conducted a series of rigorous tests using MIUI community data to evaluate the performance of our predictive model. The results show that our EEVC-based idea-screening model achieves state-of-the-art performance, outperforming the 3Cs model.

To assess the robustness of the model, we further tested it on a new MIUI dataset and a dataset collected from the Huawei Product Design Community. The results consistently show that our model outperforms the 3Cs baseline in both datasets. We then employed the SHAP module to analyze the contribution of each feature and found that the EEVC components play a vital role in idea screening. In particular, the exploration and exploitation of digital resources—as part of the idea's co-creation context—proved to be especially critical. Finally, we constructed a quasi-optimal feature subset for the best-performing model, consisting of the top 35 features ranked by SHAP values.

7.1. Implications for research

First, we propose, implement, and evaluate a theory-driven predictive framework to address idea-screening challenges in collaborative crowdsourcing for open innovation. Automated idea screening can help overcome information overload and prediction challenges [11]. Although there have been pioneering developments in idea-screening systems (e.g., [4,40]), these systems still fail to capture important aspects of open innovation communities and their ideas. Our research employs the design science research paradigm to propose a high-performance design framework that automates the fuzzy front-end stage of NPD. From the EEVC perspective, we construct a comprehensive yet parsimonious framework for idea screening. Our design evaluations demonstrate the superior performance of the EEVC framework and the relative importance of each theoretical dimension. Our work contributes to the NPD and idea screening literature by providing a novel theory-driven idea-screening framework, specifically, the EEVC-based interpretable framework for machine learning.

Second, we propose and validate the cumulative digital resources within a community as a new and significant theoretical dimension for predicting the likelihood of idea implementation. The 3Cs model by Hoornaert et al. [4] combines the content, contributor, and crowd perspectives to assess their predictive validity for idea implementation.

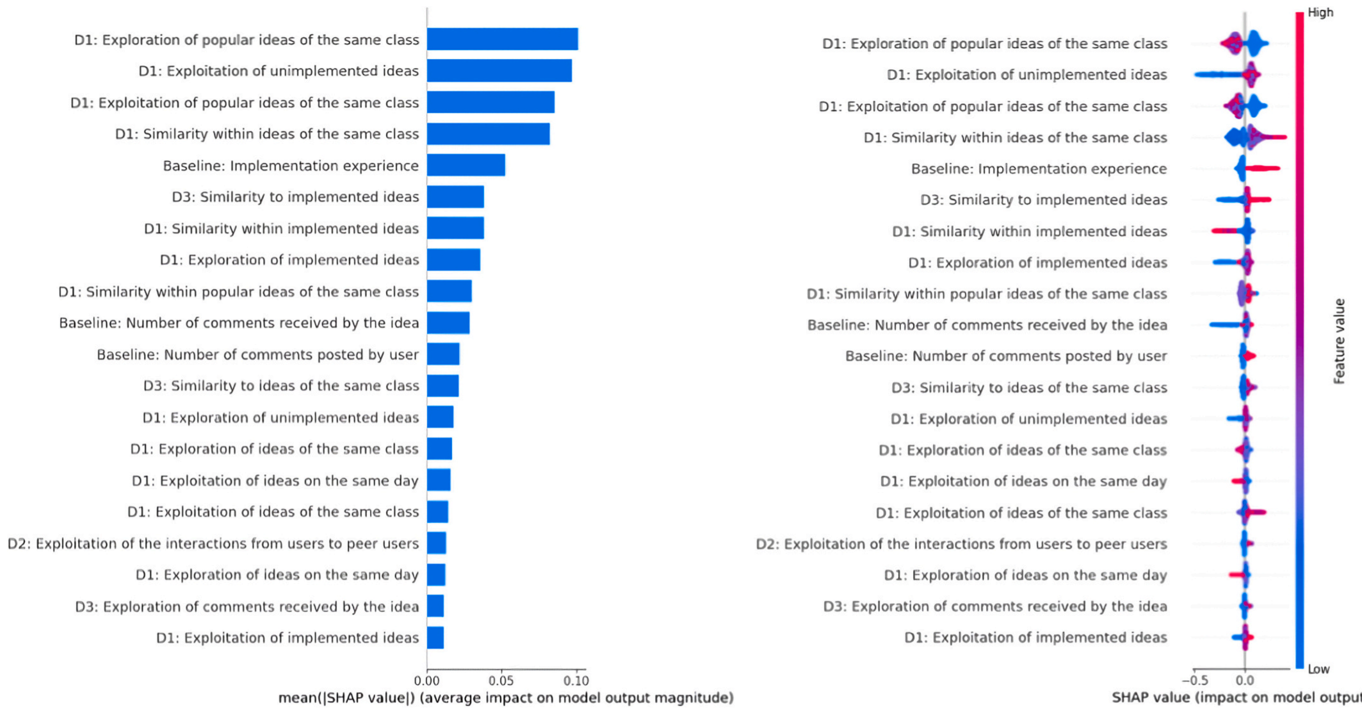

Fig. 4. Top 20 features in terms of SHAP values.

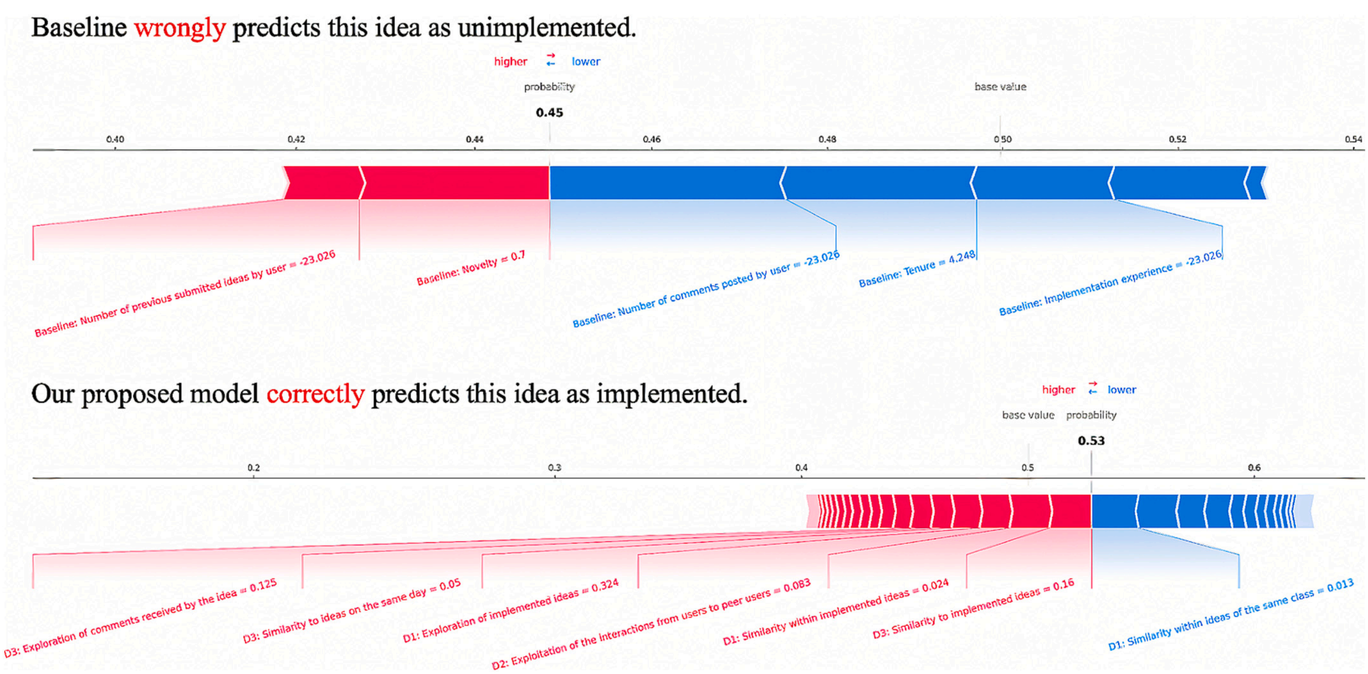

Fig. 5. An example of idea screening.

Unlike the integrative approach of the 3Cs model, we conceptualize open innovation for collaborative crowdsourcing systems as a value cocreation model [18] that involves three types of actors—platform systems, users, and firm experts—and emphasizes their collaborative interactions and creative outcomes. Based on the fluidity and knowledge evolution mechanisms within collaborative crowdsourcing communities [19], we propose that the accumulated digital resources serve as the context for subsequent value cocreation activities. Accordingly, we define three analytical levels within the value cocreation system: the creative context, direct interactions, and creative outcomes. For both the 3Cs model and Grönroos' [18] value cocreation model, digital resources as a creative context offer a novel analytical perspective. Moreover,

Table 9
Model performance by feature selection.

	Task 1		Task 2 (balanced)		Task 2 (unbalanced)	
	With Top 35 Features	Full Model	With Top 35 Features	Full Model	With Top 35 Features	Full Model
Acc.	0.741	0.729	0.708	0.690	0.712	0.691
Pre.	0.716	0.708	0.707	0.698	0.557	0.543
Rec.	0.805	0.794	0.709	0.690	0.708	0.691
F1	0.763	0.749	0.708	0.687	0.487	0.462
AUC	0.829	0.808	0.783	0.762	0.774	0.762
SRC	0.589	0.526	0.498	0.454	0.254	0.211
NDCG	0.974	0.962	0.972	0.955	0.820	0.767

Note 1. Pre. = precision; Rec. = recall; F1==F1 score; Acc. = accuracy.
Note 2. The dataset from Scenario 3 is used for Task 2 to generate the results.

SHAP value analysis indicated that the exploration and exploitation of digital resources significantly increase the predictive accuracy of the likelihood of idea implementation. In particular, SHAP value analysis revealed that 13 of the top 20 features contributing to predictive accuracy stem from the exploration and exploitation of digital resources. This digital resource dimension provides a new perspective on the idea-screening problem and extends Grönroos' [18] value cocreation model.

Third, to the best of our knowledge, we are the first to apply exploration and exploitation to predict the likelihood of idea implementation, advancing the conceptualization and operationalization of the EEVC theoretical framework. To balance short-term economic returns with long-term competitive advantage, firms should comprehensively evaluate initial NPD ideas [5,1]. Idea implementation serves as a holistic metric for assessing the novelty, feasibility, and commercial value of ideas [21,22]. Expanding March's [20] exploration and exploitation framework, we propose that a value cocreation system incorporating both the exploration of new possibilities and the exploitation of established certainties is more likely to generate ideas that meet comprehensive implementation standards. Under this theoretical framework, we developed six theoretical dimensions: the exploration or exploitation of digital resources, the exploration or exploitation of direct interactions, and the exploration or exploitation of ideas and their comments. Based on these six dimensions, we constructed a set of predictive factors and designed measurement metrics for predicting idea implementation (see Table 3). The conceptualization and operationalization of the EEVC framework broaden the theoretical understanding of the value cocreation model and idea implementation, laying a foundation for future empirical research.

7.2. Implications for practice

We recommend that firms hosting collaborative crowdsourcing communities establish an idea-screening workflow that integrates a machine learning system with expert evaluations. Specifically, this workflow could involve a sequence where the machine learning system predicts the likelihood of idea implementation, generates a recommended set of ideas with high implementation potential, and then allows firm experts to assess these ideas to make final implementation decisions. The machine learning system can automatically filter out redundant, low-quality, or fake submissions while predicting implementation probabilities and accurately recommending high-potential ideas. In the context of uncertain success for future products or features, combining machine intelligence with human judgment offers a cost-effective approach that addresses the prediction challenges in idea screening by leveraging the respective strengths of AI and human expertise.

The EEVC-based idea-screening system also provides an additional advantage by supporting digital resource management within communities [19]. Our research highlights the importance of digital resource management both theoretically and empirically, and we offer several implementation suggestions. For example, communities could prioritize displaying ideas, whether implemented or not, that exhibit significant exploratory or exploitative value. For specific task categories, communities could identify clusters of highly similar ideas, showcasing the problems that these ideas address and the evolution of related discussions. Furthermore, communities could promote idea clusters within the same category that possess substantial exploratory or exploitative value. Alternatively, communities could limit the visibility of ideas submitted on the same day to avoid potential imitation and competitive tensions among users.

To implement an EEVC-based idea-screening system in open innovation communities, we suggest that practitioners follow five steps: data collection, feature extraction, model building, system integration, and feedback optimization. First, historical community data should be collected and stored in an outcome repository and an idea repository, which will serve as the source for foundational data in subsequent feature extraction and analysis. Second, feature sets should be constructed via the EEVC model and the 3Cs model, and a range of analytical methods (refer to Section 4.2) should be used to ensure comprehensive and accurate feature extraction. Third, the extracted features should be applied to model training, and cross-validation should be employed to create training and testing datasets for building the EEVC idea implementation prediction model. Fourth, the EEVC prediction model should be integrated into the open innovation platform. This integration will enable a real-time interface between idea submission and screening workflows and allow the system to automatically screen submitted ideas and recommend high-potential ideas to expert panels. Finally, the screening criteria and model parameters should be refined based on expert feedback, continuously optimizing the system's recommendation capabilities to better align with expert needs.

7.3. Limitations and future work

Several limitations to this study merit discussion and create opportunities for future research. First, owing to data limitations, this research does not consider the unobservable group decision-making process of an internal support system consisting of firm experts, which may be an important factor affecting idea implementation [75,76]. Future research should focus on group decision-making by firm experts and explore its causes and consequences.

Second, this study validated the EEVC idea-screening model exclusively within Xiaomi's MIUI community and the Huawei Product Design Community, two representative open innovation environments within the Chinese context. While these settings offer valuable insights, they also raise questions about the applicability of the EEVC model in different cultural, linguistic, and community-specific contexts. Future research could examine the model's portability to Western open innovation communities, such as Dell's IdeaStorm or Starbucks' My Starbucks Idea, to evaluate its effectiveness in English-speaking settings.

Third, this study uses commonly adopted machine learning algorithms (e.g., the AdaBoost, SVM, CNN, and XGBoost algorithms) to validate the proposed EEVC-based framework. While effective, this approach lacks methodological innovation. Future research could focus on developing novel machine learning algorithms that more deeply integrate the EEVC framework. These advancements would enhance methodological rigor and expand the framework's applicability to more complex idea-screening tasks.

Another interesting theme for future research is the long-term balance between selecting the efficiency and diversity of user ideas. The idea-screening model potentially limits the preferences of firm experts to existing interests [77]. Future research can explore the allocation of decision-making power between idea-screening tools and firm experts to find feasible measures that enhance both screening efficiency and the diversity of ideas.

CRediT authorship contribution statement

Qian Liu: Writing – review & editing, Writing – original draft, Conceptualization. **Qianzhou Du:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Chuang Tang:** Writing – review & editing, Formal analysis. **Yili Hong:** Writing – review & editing, Writing – original draft, Conceptualization. **Weiguo Fan:** Writing – review & editing, Project administration, Conceptualization.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Qianzhou Du reports financial support was provided by National Natural Science Foundation of China. Qian Liu reports financial support was provided by National Natural Science Foundation of China. Qian Liu reports financial support was provided by Beijing Natural Science Foundation. Qian Liu reports financial support was provided by Double First-class construction project of Central University of Finance and Economics. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Appendix

Latent Dirichlet Allocation

We applied the latent Dirichlet allocation (LDA) model [59] to idea content to discover the main topics related to customer requirements. The LDA model strives to extract a hidden structure that is based on the co-occurrence of observed words in different documents and consists of a set of topics in the whole body of text. To determine the topic number k, we follow Griffiths and Steyvers [62] and calculate the coherence scores of LDA models with different topic numbers $k \in K = \{5, 6, 7, \ldots, 25\}$. Based on Fig. I, we set $k = 14$ as the best topic number, i.e., the number for which the LDA model has the highest coherence score. The LDA model with $k = 14$ calculates the posterior distribution of the unobserved variables in the collection of documents. Given a set of documents, the LDA's output contains two main components: a list of topics associated with a set of words, as shown in Table I, and a list of documents with a vector, each dimension of which indicates the probability of a document belonging to a specific topic. We define topic dispersion by using the Gini index of the probabilities associated with all topics. A high score for topic dispersion means that the content is likely to focus on one or a few topics. Smaller scores indicate that the idea content covers many topics. In addition to the Gini index, we adopted the standard deviation, the Herfindahl–Hirschman index, and entropy to quantify topic dispersion. The results show that the Gini index performs best, although no significant improvement is obtained versus the other three measurements.

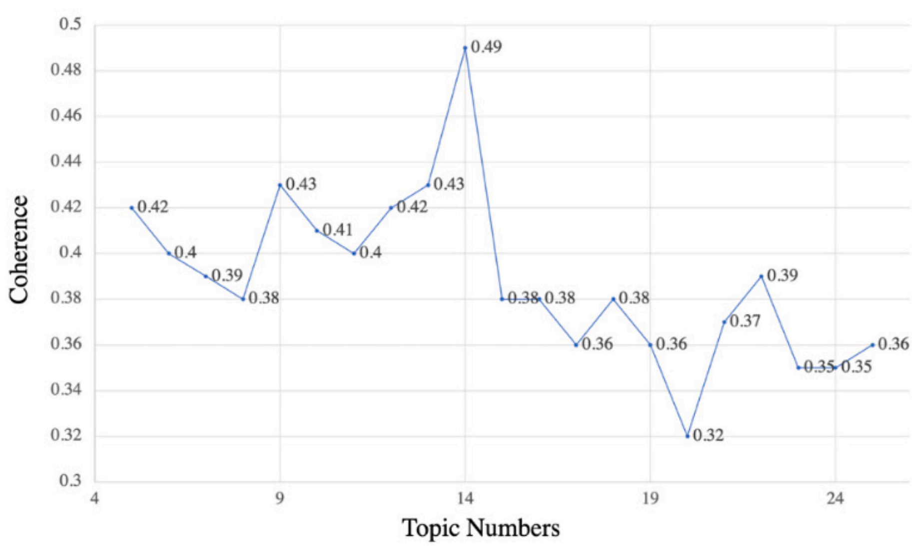

Fig. I. Coherence scores of LDA models with different topic numbers.

Table I
List of topics identified using latent Dirichlet allocation.

Topic 1 Software features	Topic 2 Music playback	Topic 3 Theme mode
Function	Music	Mode
Mobile phone	File	Image
System	QQ	Stuff
Software	SMS	Single hand
Version	Interface	Apple
User	Memory	Wallpaper
Advertising	App	Permissions
Time	Bring your own	Lyrics
Voice	Internet	Button
Split screen	Situation	Effect
Topic 4 User interaction	Topic 5 System features	Topic 6 Icon interface
Screen	Topic	Icon
Flow rate	Forum	Desktop
Phone	Font	Status bar
Gesture	Information	Folder
Shutdown	Weather	State
MIUI fans	Contact	Procedure
Call	Computer	Velocity
Space	Hour	Desktop icon
Habit	Firm	Page
Backup	Call logs	Method
Topic 7 Video playback	Topic 8 Password security	Topic 9 Voice assistant
Video	Password	Aides
Turn on	Message	Center
Bluetooth	China Unicom	Voice
Humanized	Encrypt	Black screen
Distance	Models	It's too big
Frequency	China Telecom	Team
Drawing	Samsung	Products
Chat	Title	Wireless
Distinguish	Head portrait	Pop-ups
Shield	Standard	Router
Topic 10 Photo browsing	Topic 11 Game experience	Topic 12 Power life
Photo	Keystroke	Power supply
Volume	Game	Post
Signal	Engineer	Be careful
Performance	Brightness	Weather forecast
Earphone	Operators	Think about it
Media	Physics	Order
Machine	Level	Mail
View	Mobile phone number	Type
Target	Percentage	Quantity
Loudspeaker	Definition	Life span
Topic 13 System version	Topic 14 Mobile hotspots	
Stable version	Hot spot	
Battery	Ask	
Electricity	Got it	
Essential	Vote	
Standby	Phone cards	
Environment	Standard version	
Developer	Family	
Selectivity	Number	
Example	Slow motion	
Power consumption	Big screen	

Table II
Summary statistics of the MIUI NFD forum.

Dataset	MIUI NFD forum
No. of ideas	89,906
No. of ideators	53,885
No. of users	201,655
Average no. of replies per idea	7.78
No. of words in the dataset	13,366,946

Table III
Coding rules for idea implementation.

Coding rules	Seal types	Definition
Ideas are implemented by firm experts (labeled “1”)	SUBMITTED	The idea has been adopted by firm experts and submitted to the product R&D group.
	COLLECTION	The idea has been adopted by firm experts and collected by the product R&D group.
	PROCESSING	The product R&D group is launching an idea-related development.
	RESOLVED	The product R&D group has completed product updates related to the idea.
Firm experts did not implement the idea (labeled “0”)	No seal	Ideas that did not receive the firm experts’ seal were not implemented.
	UNRESOLVED	The idea has not been adopted or implemented.
	PLEASE ADD	The idea lacks adequate information.
	TO BE DISCUSSED	The idea needs further discussion and will not be implemented for the moment.
	HAVE A REPLY	Firm experts have responded to the issues involved with the idea.

Table IV
Summary statistics of the outcome of ideas.

	Total	Implemented	Not implemented
2010	1599	126	1473
2011	13,682	245	13,437
2012	12,997	1254	11,743
2013	11,358	833	10,525
2014	18,078	1908	16,170
2015	32,329	1852	30,477
Total	89,906	6218	83,688

Table V
Measurement of features in the EEVC-based idea screening model.

Theoretical dimensions	Features	Measure	Method
D1: Exploration/exploitation of digital resources	Exploration of implemented ideas	Exploration score of implemented ideas in the community before submitting idea *i*	Lexicon-based
	Exploitation of implemented ideas	Exploitation score of implemented ideas in the community before submitting idea *i*	
	Similarity within implemented ideas	Degree of similarity within implemented ideas in the community before submitting idea *i*; lower similarities represent exploration, whereas higher similarities represent exploitation	Similarity-based
	Exploration of unimplemented ideas	Exploration score of implemented ideas in the community before submitting idea *i*	Lexicon-based
	Exploitation of unimplemented ideas	Exploitation score of implemented ideas in the community before submitting idea *i*	
	Similarity within unimplemented ideas	Degree of similarity within unimplemented ideas in the community before submitting idea *i*; lower similarities represent exploration, whereas higher similarities represent exploitation	Similarity-based
	Exploration of ideas of the same class	Exploration score of same-class ideas in the community before submitting idea *i*	Lexicon-based
	Exploitation of ideas of the same class	Exploitation score of same-class ideas in the community before submitting idea *i*	
	Similarity within ideas of the same class	Degree of similarity within same-class ideas in the community before submitting idea *i*; lower similarities represent exploration, whereas higher similarities represent exploitation	Similarity-based
	Exploration of popular ideas of the same class	Exploration score of popular ideas of the same class in the community before submitting idea *i*	Lexicon-based
	Exploitation of popular ideas of the same class	Exploitation score of popular ideas of the same class in the community before submitting idea *i*	
	Similarity within popular ideas of the same class	Degree of similarity within popular ideas of the same class in the community before submitting idea *i*; lower similarities represent exploration, whereas higher similarities represent exploitation	Similarity-based
	Exploration of ideas on the same day	Exploration score of same-day ideas in the community before submitting idea *i*	Lexicon-based
	Exploitation of ideas on the same day	Exploitation score of same-day ideas in the community before submitting idea *i*	
	Similarity within ideas on the same day	Degree of similarity within same-day ideas in the community before submitting idea *i*; lower similarities represent exploration, whereas higher similarities represent exploitation	Similarity-based
D2: Exploration/exploitation of direct interactions	Degree of involvement in idea classes.	User’s degree of involvement in idea *i* classes; lower involvement represents exploration, whereas higher involvement represents exploitation	Statistics-based

(*continued on next page*)

Table V (*continued*)

Theoretical dimensions	Features	Measure	Method
	Exploration of the interactions between users and firm experts	Exploration score of comments from firm experts on the user's previous ideas before submitting idea *i*	Lexicon-based
	Exploitation of the interactions between users and firm experts	Exploitation score of comments from firm experts on the user's previous ideas before submitting idea *i*	
	Exploration of the interactions from users to peer users	Exploration score of user's comments to peer users' ideas before submitting idea *i*	
	Exploitation of the interactions from users to peer users	Exploitation score of user's comments to peer users' ideas before submitting idea *i*	
	Exploration of the interactions from peer users to users	Exploration score of comments from peer users on the user's previous ideas before submitting idea *i*	
	Exploitation of the interactions from peer users to users	Exploitation score of comments from peer users on the user's previous ideas before submitting idea *i*	
D3: Exploration/exploitation of ideas and their comments	Similarity to implemented ideas	Degree of similarity of idea *i* to implemented ideas in the community; lower similarities represent exploration, whereas higher similarities represent exploitation	Similarity-based
	Similarity to unimplemented ideas	Degree of similarity of idea *i* to unimplemented ideas in the community; lower similarities represent exploration, whereas higher similarities represent exploitation	
	Similarity to ideas of the same class	Degree of similarity of idea *i* to same-class ideas in the community; lower similarities represent exploration, whereas higher similarities represent exploitation	
	Similarity to popular ideas of the same class	Degree of similarity of idea *i* to popular ideas of the same class in the community; lower similarities represent exploration, whereas higher similarities represent exploitation	
	Similarity to ideas on the same day	Degree of similarity of idea *i* to same day ideas in the community; lower similarities represent exploration, whereas higher similarities represent exploitation	
	Exploration of comments received by the idea	Exploration score of comments received by idea *i*	Lexicon-based
	Exploitation of comments received by the idea	Exploitation score of comments received by idea *i*	

Table VI
Idea categories.

	Category	Frequency	Percentage
1	Unknown	20,134	22.36
2	Other	10,939	12.15
3	System settings	5119	5.68
4	Desktop	4864	5.40
5	Status bar	4008	4.45
6	Lock screen	3483	3.86
7	System update	3393	3.76
8	Security center	3007	3.33
9	Music	2913	3.23
10	Xiaomi cloud service	2863	3.17
11	Camera	2552	2.83
12	Call	2235	2.48
13	Gallery	2182	2.42
14	Dial	2072	2.30
15	SMS	2028	2.25
16	Contact	1946	2.16
17	Theme	1692	1.87
18	File manager	1401	1.55
19	Browser	1397	1.55
20	Calendar	1203	1.33
21	Weather	1062	1.17
22	Clock or Calculator	1061	1.17
23	Post-it notes or Compass	946	1.05
24	DND model	865	0.96
25	Video	830	0.92
26	Yellow pages of life	784	0.87
27	Voice assistant	650	0.72
28	App store	615	0.68
29	Xiaomi account	494	0.54
30	*E*-mail	390	0.43
31	Tape recorder	310	0.34
32	Xiaomi life	310	0.34
33	Password protection	289	0.32
34	Download manager	261	0.28

(*continued on next page*)

Table VI (*continued*)

	Category	Frequency	Percentage
35	Custom input method	261	0.28
36	Broadcasting station	254	0.28
37	Radio	247	0.27
38	Xiaomi Wallet	212	0.23
39	Backup	183	0.20
40	Game center	166	0.18
41	Levels & Quests	96	0.10
42	Xiaomi	95	0.10
43	Xiaomi Financial	93	0.10
44	QR Code Scanning	56	0.06
45	Xiaomi Fast Pass	37	0.04
46	Xiaomi Card Package	32	0.03
47	Smart home	13	0.01
		90,043	100

Table VII
Evaluation metrics.

Evaluation Metrics	Equation
Accuracy	$\boldsymbol{accuracy} = \frac{\text{number of correctly predicted ideas}}{\text{number of predicted ideas}}$,
Precision	$\boldsymbol{precision} = \frac{\text{number of correctly predicted positive ideas}}{\text{number of predicted positive ideas}}$,
Recall	$\boldsymbol{recall} = \frac{\text{number of correctly predicted positive ideas}}{\text{number of actual positive ideas}}$,
F1 score	$\textbf{F1 score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$,
AUC	$\boldsymbol{AUC}$ = area under the curve plots (TPR, FPR), where TPR (true positive rate) $= \textit{recall}$ and FPR (false positive rate) $= \frac{\text{number of wrongly predicted positive ideas}}{\text{number of actual negative ideas}}$,
SRC	$\boldsymbol{SRC}$ (Spearman's rank correlation) $= \rho = 1 - \frac{6\sum {d_i}^2}{n(n^2-1)}$, where d_i = difference in paired ranks and n = number of cases,
NDCG	$\boldsymbol{NDCG} = \frac{DCG_{pos}}{iDCG}$, where $iDCG$ is the ideal value of DCG, and $DCG_{pos} = \sum_{pos=1}^{N} \frac{\text{relevance}_{pos}}{\ln(pos+1)}$.

Table VIII
Results of the NDCG for Task 1.

Feature Sets	CNN	AdaBoost	Logistic Regression	SVM	Random Forest	XGBoost
Baseline	0.896	0.932	0.934	0.933	0.933	0.934
	(0.014)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Baseline + D3	0.918*	0.943*	0.930	0.941*	0.946**	0.950***
	(0.011)	(0.004)	(0.008)	(0.006)	(0.008)	(0.006)
Baseline + D2	0.918*	0.936	0.937	0.933	0.942*	0.945*
	(0.009)	(0.010)	(0.007)	(0.007)	(0.005)	(0.006)
Baseline + D1	0.921**	0.948**	0.935	0.944**	0.945**	0.947**
	(0.011)	(0.007)	(0.008)	(0.006)	(0.006)	(0.005)
Baseline + D1 + D2 + D3	**0.925****	**0.957*****	**0.941**	**0.946****	**0.958*****	**0.962*****
	(0.015)	(0.005)	(0.005)	(0.005)	(0.006)	(0.004)

Three additional baseline models

To further demonstrate the superiority of the proposed EEVC-based idea screening model, we incorporated three additional baseline models drawn from recent high-quality studies [43,45,13], each recognized for their strong performance. These models were selected for their close alignment with the core themes of idea screening and crowd selection in online platforms, thus providing a robust benchmark for comparison.

Qin et al. [68] developed a comprehensive feature set using data from the Xiaomi MIUI forum, integrating both idea-level quantitative indicators and user characteristics (baseline 1). Their framework categorized idea attributes into four key dimensions—community recognition, content quality, idea semantics, and user attitude—to predict idea adoption outcomes (see Table IX). Similarly, Dahlander et al. [69] constructed their model based on data from the LEGO Ideas platform, emphasizing user-centric features (Baseline 2). They defined their features across six dimensions—user attributes, user experience, user status, idea topics, and idea-related image and text characteristics—to assess the level of community support for each idea (see Table IX).

In addition, Hsiang and Rayz [70] approached idea evaluation from a semantic perspective (Baseline 3). By using multiple text representation techniques and a range of supervised machine learning models, they extracted semantic information to predict contributor popularity on the My Starbucks Idea platform. Their empirical results showed that the TF-IDF representation yielded the best performance among the evaluated methods. Accordingly, we adopted the TF-IDF approach in our own semantic processing tasks (see Table X). Although their primary focus was on contributor popularity rather than idea quality, their methodological approach to identifying valuable user-generated content remains highly relevant to our research objective of distinguishing high-quality ideas.

Given the distinct but complementary emphases of these three studies, benchmarking our proposed model against them enables a more comprehensive and rigorous evaluation of its predictive power in idea implementation. To ensure consistency, we followed the same evaluation procedure as in Task 1, thus extending the results originally reported in Table 6, as shown in Table XI. However, owing to platform-specific differences between the datasets used in the original studies and our Xiaomi MIUI forum dataset, several features were either unavailable or inaccessible in our context. As a result, full replication of these feature sets was not feasible. Nonetheless, we made every effort to approximate their representation based on the available characteristics of our dataset. Detailed information regarding this feature adaptation is provided in Table IX and Table X.

Table IX
Feature set of baselines 1 and 2.

Variable Type	Variable Name	Definition	Baseline 1 [68]	Baseline 2 [69]
User attributes	Tenure	Number of days between the user profile was created and idea submission	–	✓
	Age	Age of the user measured in years	–	–
	Gender	Gender of the user	–	–
User attitude	Users' self-presentation	Whether the user's name is modified for himself; 1 if yes, 0 if no	✓	–
User experience	Ideas submitted	Number of ideas submitted before the focal idea of the focal user	–	✓
	Previous success	Proportion of the user's previously submitted ideas that received official adoption before the submission of the focal idea	–	✓
	Comments written	Number of comments written on others' ideas before submission of the focal idea	–	✓
	Sentiment of comments	Sentiment of *Comments written* based on the dictionary of NTUSD [78]	–	✓
	Variation in sentiment	The average of the standard deviations of *Sentiment of comments*	–	✓
	Logins	Number of logins up before submission of the focal idea	–	–
	Votes cast	Number of supports the focal user has given before submission of the focal idea	–	–
	Users followed	Number of users the focal user was following before submission of the focal idea	–	–
User status	Total votes received	Number of credits the focal user received before submission of the focal idea	–	✓
	Highest number of votes received	Maximum number of credits the focal user received on a single idea before submission of the focal idea	–	✓
	Comments received	Number of comments received before submission of the focal idea	–	✓
	Sentiment of comments received	sentiment of *Comments received* before submission of the focal idea based on the dictionary of NTUSD [78]	–	✓
	Variation in sentiment received	Average of the standard deviations of *Sentiment of comments* received	–	✓
	Personal followers	Maximum number of followers the focal user has had on the submission dates of former projects	–	–
	Unique followers	Number of users who were following the focal user before submission of the focal idea	–	–
Idea topics	Idea topics	Topic categories extracted using LDA from project titles, descriptions, and metadata	–	–
	Number of tags	Number of tags assigned to the idea	–	–
Idea semantics	Number of idea themes	The number of extracted keywords included in the idea	✓	–
	Number of images	Number of images uploaded with the idea	–	✓
	Size of the principal image	The original size of the principal image (in kilobytes)	–	–
	Colorfulness	Calculated a measure for the diversity of color usage in the idea's principal image with the OpenCV Python package [79]	–	–
	Complexity	Measured the complexity by calculating the principal image's relationship between file size (in kilobytes) to area (in square inches)	–	–
Idea text characteristics	Length of description	The number of characters in an idea, including text, punctuation marks, and special symbols	✓	✓
	Sentiment of description	Sentiment of the focal idea description based on the dictionary of NTUSD [78]	–	✓
	Resubmission count	Number of idea resubmissions before the submission of the focal idea	–	–
Content quality	Idea richness	Whether images are included in the idea; 1 if yes, 0 if no	✓	–
	Timeliness	The number of days between the day crawled data when the idea is published	–	–
Community recognition	Number of likes	The number of likes the idea has received from other users in the community	✓	–
	Number of comments	The number of comments the idea has received from other users in the community	✓	–
	Number of related posts	Number of idea posts included under the same proposal	–	–

Note 1. The features actually utilized in the extended experiments are marked with a "✓", whereas those not included are indicated with a "–".

Table X
Model settings of baseline 3 [70].

Classifier	Text Representation	Baseline 3 [70]
Random Forest	Count vector	–
Random Forest	TF-IDF representation	✓
Random Forest	Word embedding	–
XGBoost	Count vector	–
XGBoost	TF-IDF representation	✓
XGBoost	Word embedding	–

Table XI
Evaluation results of task 1.

Feature Sets	CNN			AdaBoost			Logistic Regression			SVM			Random Forest			XGBoost		
	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1
Baseline 1 [68]	0.573	0.509	0.457	0.549	0.647	0.594	0.573	0.259	0.356	0.531	0.587	0.557	0.535	0.553	0.544	0.540	0.574	0.556
	(0.027)	(0.097)	(0.065)	(0.004)	(0.023)	(0.009)	(0.009)	(0.016)	(0.015)	(0.004)	(0.011)	(0.005)	(0.005)	(0.013)	(0.008)	(0.006)	(0.018)	(0.011)
Baseline 2 [69]	0.611	0.502	0.527	0.602	0.561	0.580	0.607	0.495	0.545	0.589	0.504	0.543	0.579	0.609	0.593	0.581	0.570	0.575
	(0.019)	(0.056)	(0.026)	(0.004)	(0.009)	(0.006)	(0.003)	(0.009)	(0.005)	(0.005)	(0.003)	(0.001)	(0.005)	(0.008)	(0.006)	(0.008)	(0.010)	(0.009)
Baseline 3 [70]	0.590	0.575	0.581	0.590	0.508	0.543	0.610	0.621	0.615	0.616	0.625	0.621	0.603	0.609	0.606	0.593	0.582	0.587
	(0.017)	(0.048)	(0.027)	(0.026)	(0.070)	(0.035)	(0.016)	(0.022)	(0.016)	(0.015)	(0.016)	(0.013)	(0.015)	(0.015)	(0.013)	(0.016)	(0.025)	(0.017)
Baseline 4 (3Cs)	0.618	0.649	0.632	0.630	0.684	0.653	0.621	0.667	0.643	0.621	0.685	0.651	0.625	0.662	0.643	0.630	0.670	0.650
	(0.019)	(0.032)	(0.017)	(0.017)	(0.018)	(0.013)	(0.017)	(0.019)	(0.014)	(0.017)	(0.020)	(0.014)	(0.018)	(0.017)	(0.014)	(0.017)	(0.019)	(0.015)
Baseline 4 + D3	0.621	0.658	0.639	0.649*	0.709**	0.678**	0.655***	0.690**	0.672***	0.653***	0.744***	0.695***	0.649**	0.753***	0.696***	0.663***	0.748***	0.702***
	(0.021)	(0.091)	(0.042)	(0.014)	(0.015)	(0.011)	(0.019)	(0.022)	(0.018)	(0.016)	(0.019)	(0.014)	(0.022)	(0.023)	(0.014)	(0.015)	(0.018)	(0.012)
Baseline 4 + D2	0.619	0.654	0.636	0.633	0.684	0.657	0.635	0.668	0.651*	0.637*	0.702*	0.667*	0.632	0.728***	0.676***	0.645*	0.724***	0.682***
	(0.020)	(0.082)	(0.033)	(0.024)	(0.022)	(0.017)	(0.022)	(0.020)	(0.015)	(0.018)	(0.025)	(0.013)	(0.018)	(0.025)	(0.016)	(0.018)	(0.017)	(0.014)
Baseline 4 + D1	0.626*	0.701***	0.661***	0.642*	0.703**	0.671**	0.650**	0.708***	0.677***	0.658***	0.759***	0.705***	0.659***	0.776***	0.712***	0.668**	0.751***	0.707***
	(0.022)	(0.068)	(0.024)	(0.020)	(0.016)	(0.014)	(0.021)	(0.024)	(0.018)	(0.020)	(0.018)	(0.013)	(0.019)	(0.018)	(0.013)	(0.013)	(0.019)	(0.007)
Baseline 4 + D1 + D2 + D3	0.634**	0.716***	0.673***	0.679***	0.728***	0.703***	0.674***	0.727***	0.699***	0.676***	0.782**	0.725***	0.685***	0.790***	0.734***	0.708***	0.794***	0.749***
	(0.023)	(0.037)	(0.021)	(0.016)	(0.015)	(0.009)	(0.018)	(0.019)	(0.014)	(0.019)	(0.018)	(0.012)	(0.016)	(0.013)	(0.010)	(0.013)	(0.017)	(0.006)
	Acc.	AUC	SRC	Acc.	AUC	SRC	Acc.	AUC	SRC	Acc.	AUC	SRC	Acc.	AUC	SRC	Acc.	AUC	SRC
Baseline 1 [68]	0.537	0.562	0.107	0.558	0.588	0.152	0.532	0.554	0.094	0.534	0.548	0.084	0.536	0.550	0.087	0.542	0.563	0.110
	(0.007)	(0.012)	(0.020)	(0.004)	(0.005)	(0.008)	(0.003)	(0.004)	(0.007)	(0.005)	(0.005)	(0.008)	(0.006)	(0.006)	(0.010)	(0.007)	(0.007)	(0.013)
Baseline 2 [69]	0.578	0.624	0.216	0.595	0.642	0.245	0.587	0.626	0.219	0.576	0.618	0.204	0.583	0.621	0.209	0.579	0.615	0.199
	(0.008)	(0.010)	(0.018)	(0.004)	(0.005)	(0.008)	(0.002)	(0.003)	(0.006)	(0.003)	(0.004)	(0.007)	(0.005)	(0.007)	(0.012)	(0.008)	(0.009)	(0.015)
Baseline 3 [70]	0.587	0.623	0.212	0.576	0.603	0.182	0.612	0.66	0.277	0.618	0.662	0.281	0.604	0.648	0.256	0.591	0.625	0.217
	(0.016)	(0.017)	(0.030)	(0.019)	(0.021)	(0.037)	(0.015)	(0.014)	(0.024)	(0.014)	(0.013)	(0.023)	(0.014)	(0.017)	(0.030)	(0.015)	(0.020)	(0.034)
Baseline 4 (3Cs)	0.623	0.574	0.127	0.638	0.691	0.330	0.631	0.685	0.321	0.633	0.687	0.324	0.633	0.691	0.330	0.638	0.698	0.343
	(0.017)	(0.037)	(0.065)	(0.013)	(0.014)	(0.025)	(0.013)	(0.015)	(0.025)	(0.012)	(0.015)	(0.026)	(0.014)	(0.015)	(0.026)	(0.014)	(0.015)	(0.026)
Baseline 4 + D3	0.625	0.637***	0.237***	0.661***	0.726***	0.390***	0.661***	0.715***	0.372***	0.672**	0.731***	0.400***	0.670***	0.734***	0.406***	0.681***	0.749***	0.431***
	(0.010)	(0.011)	(0.018)	(0.009)	(0.011)	(0.018)	(0.015)	(0.013)	(0.022)	(0.010)	(0.009)	(0.015)	(0.014)	(0.014)	(0.024)	(0.010)	(0.010)	(0.017)
Baseline 4 + D2	0.592	0.638***	0.238***	0.641	0.700	0.347*	0.639	0.695	0.338	0.648	0.704	0.354	0.649*	0.711*	0.365*	0.657**	0.722**	0.384**
	(0.009)	(0.009)	(0.016)	(0.015)	(0.017)	(0.029)	(0.014)	(0.013)	(0.023)	(0.011)	(0.008)	(0.013)	(0.012)	(0.013)	(0.023)	(0.012)	(0.013)	(0.022)
Baseline 4 + D1	0.627	0.674***	0.300***	0.654*	0.722***	0.383***	0.663***	0.722***	0.385***	0.687***	0.744***	0.422***	0.696***	0.756***	0.444***	0.671***	0.741***	0.418***
	(0.016)	(0.017)	(0.030)	(0.012)	(0.015)	(0.027)	(0.016)	(0.014)	(0.024)	(0.013)	(0.012)	(0.020)	(0.010)	(0.010)	(0.017)	(0.006)	(0.004)	(0.007)
Baseline 4 + D1 + D2 + D3	0.655*	0.686***	0.317***	0.691***	0.761***	0.452***	0.685***	0.753***	0.428***	0.703***	0.762***	0.449***	0.717***	0.791***	0.509***	0.729***	0.808***	0.526***
	(0.014)	(0.019)	(0.024)	(0.007)	(0.011)	(0.019)	(0.010)	(0.009)	(0.016)	(0.010)	(0.008)	(0.014)	(0.008)	(0.011)	(0.015)	(0.006)	(0.007)	(0.013)

Note 1. Pre. = precision; Rec. = recall; F1=F1 score; Acc. = accuracy.
Note 2. Significant results compared with the baseline model are highlighted in bold, where * indicates $p < 0.1$; ** indicates $p < 0.05$; and *** indicates $p < 0.01$.
Note 3. The number in parentheses is the standard deviation of model performance in terms of the given metric.
Note 4. D1 is the exploration and exploitation of digital resources; D2 is the exploration and exploitation of direct interactions; and D3 is the exploration and exploitation of ideas and their comments.
Note 5. Baseline 1 is given by [68], Baseline 2 is given by [69], and Baselines 3 and 4 are given by [70].

Table XII. Evaluation Results of Task 2 (Scenario 1 and Scenario 2)

Table XII-1
Model performance in scenario 1.

	Scenario 1: Data-Split date is Jan. 1st, 2013									
	Balanced Testing Dataset (10 samples) (Implemented ideas = 833) (Unimplemented ideas = 833)					Unbalanced Testing Dataset (Implemented ideas = 833) (Unimplemented ideas = 10,525)				
	Baseline	Baseline + D3	Baseline + D2	Baseline + D1	Baseline + D1 + D2 + D3	Baseline	Baseline + D3	Baseline + D2	Baseline + D1	Baseline + D1 + D2 + D3
Acc.	0.512	0.526**	0.524*	0.527**	**0.551***	0.501	0.517*	0.510	0.521***	**0.540***
	(0.009)	(0.008)	(0.013)	(0.006)	**(0.010)**	(0.008)	(0.008)	(0.009)	(0.006)	**(0.008)**
Pre.	0.514	0.531**	0.526*	0.536**	**0.566***	0.493	0.505**	0.503	0.507***	**0.513***
	(0.010)	(0.010)	(0.015)	(0.010)	**(0.014)**	(0.002)	(0.003)	(0.003)	(0.003)	**(0.002)**
Rec.	0.512	0.526*	0.524*	0.527**	**0.551***	0.503	0.517*	0.510	0.521***	**0.540***
	(0.009)	(0.008)	(0.013)	(0.006)	**(0.010)**	(0.008)	(0.008)	(0.009)	(0.006)	**(0.008)**
F1	0.501	0.503	0.509	0.506	**0.545**	0.311	0.294	0.321	0.288	**0.399***
	(0.008)	(0.009)	(0.013)	(0.005)	**(0.013)**	(0.008)	(0.010)	(0.011)	(0.019)	**(0.008)**
AUC	0.515	0.538	0.537	0.541***	**0.582***	0.503	0.525	0.516*	0.533**	**0.570***
	(0.010)	(0.008)	(0.015)	(0.011)	**(0.011)**	(0.009)	(0.009)	(0.009)	(0.011)	**(0.012)**
SRC	0.051	0.066*	0.063*	0.071**	**0.142***	0.017	0.023	0.015	0.031**	**0.063***
	(0.017)	(0.014)	(0.026)	(0.018)	**(0.019)**	(0.008)	(0.008)	(0.011)	(0.009)	**(0.011)**
NDCG	0.879	0.893**	0.900***	0.893**	**0.910***	0.661	0.675**	0.677**	0.674**	**0.697***
	(0.007)	(0.005)	(0.005)	(0.009)	**(0.006)**	(0.004)	(0.005)	(0.005)	(0.004)	**(0.007)**

Table XII-2
Model performance in Scenario 2.

	Scenario 2: Data-Split date is Jan. 1st, 2014									
	Balanced Testing Dataset (10 samples) (Implemented ideas = 1908) (Unimplemented ideas = 1908)					Unbalanced Testing Dataset (Implemented ideas = 1908) (Unimplemented ideas = 16,170)				
	Baseline	Baseline + D3	Baseline + D2	Baseline + D1	Baseline + D1 + D2 + D3	Baseline	Baseline + D3	Baseline + D2	Baseline + D1	Baseline + D1 + D2 + D3
Acc.	0.527	0.543*	0.548**	0.546**	0.569***	0.537	0.544**	0.557***	0.546***	0.573***
	(0.006)	(0.010)	(0.009)	(0.006)	(0.012)	(0.006)	(0.006)	(0.020)	(0.005)	(0.007)
Pre.	0.532	0.571***	0.567***	0.561***	0.597***	0.512	0.518***	0.525***	0.521***	0.532***
	(0.007)	(0.012)	(0.011)	(0.011)	(0.018)	(0.002)	(0.002)	(0.005)	(0.003)	(0.006)
Rec.	0.524	0.543*	0.548**	0.546**	0.569***	0.527	0.544**	0.557***	0.546***	0.573***
	(0.006)	(0.011)	(0.012)	(0.006)	(0.014)	(0.006)	(0.006)	(0.020)	(0.005)	(0.007)
F1	0.518	0.507	0.513	0.521	0.586***	0.379	0.366	0.360	0.329	0.424**
	(0.007)	(0.036)	(0.044)	(0.007)	(0.038)	(0.017)	(0.014)	(0.076)	(0.014)	(0.046)
AUC	0.548	0.588***	0.596***	0.577***	0.605***	0.543	0.569**	0.597***	0.577***	0.610***
	(0.011)	(0.025)	(0.020)	(0.010)	(0.013)	(0.011)	(0.009)	(0.023)	(0.008)	(0.012)
SRC	0.099	0.152***	0.165***	0.134***	0.182***	0.059	0.074**	0.104***	0.082***	0.117***
	(0.020)	(0.044)	(0.035)	(0.018)	(0.022)	(0.011)	(0.009)	(0.025)	(0.009)	(0.013)
NDCG	0.904	0.919*	0.925***	0.915*	0.932***	0.742	0.753*	0.769***	0.759**	0.781***
	(0.006)	(0.009)	(0.006)	(0.002)	(0.007)	(0.005)	(0.006)	(0.012)	(0.003)	(0.006)

Hyper-parameter tuning of LSTM

We adopt the deep learning library PyTorch to implement the LSTM model. To achieve the best performance, we apply the grid search strategy to tune the hyper-parameter and the network construction. The details are shown below.

Table XIII
The details of LSTM's hyper-parameter tuning procedure.

Parameter Range	
Number of epochs	500
Dimension of embedding	self.embedding_dim = trial.suggest_int("embedding_dim", 10, 100, 5)
Number of layers	self.num_layers = trial.suggest_int("num_layers", 1, 5, 1)
Size of hidden layers	self.hidden_size = trial.suggest_int("hidden_size", 1, 30, 2)
Dimensions of dense layer	self.fc2_input_dim = trial.suggest_int("fc2_input_dim", 16, 128, 4)
Learning rate	trial.suggest_float("lr", 1e-4, 1e-1,log = True)
Optimizer	['SGD', 'ASGD', 'Rprop', 'Adagrad', 'Adam', 'RMSprop']
Best Hyper-parameter	

(*continued on next page*)

Table XIII (*continued*)

Parameter Range	
Task 1	{'embedding_dim': 80, 'num_layers': 3, 'hidden_size': 21, 'fc2_input_dim': 16, 'optimizer': 'Adam', 'lr': 0.00035290337904482575, 'momentum': 0.46268175520466404, 'batch_size': 48}
Task 2 - balanced	{'embedding_dim': 100, 'num_layers': 1, 'hidden_size': 21, 'fc2_input_dim': 48, 'optimizer': 'Adam', 'lr': 0.029535338404442817, 'momentum': 0.9218456503074373, 'batch_size': 32}
Task 2 - unbalanced	{'embedding_dim': 20, 'num_layers': 3, 'hidden_size': 1, 'fc2_input_dim': 80, 'optimizer': 'Adam', 'lr': 0.001835094561653265, 'momentum': 0.36836528170392013, 'batch_size': 112}

Robustness check

Since our main results were based on data before 2016, it is possible that the model might not work when applied to more recent data. To further prove the validity of our model, we collected recent data and tested our model's performance. Specifically, we scraped recent data from the MIUI NFD forum to evaluate our EEVC-based idea screening model. We obtained all the idea threads from the MIUI NFD forum from May 1, 2020, to October 15, 2024, totaling 79,531 ideas posted by 61,301 users. After removing 1380 abnormal observations (e.g., outliers and missing variables), we identified 1734 positive examples and 71,740 negative examples, indicating a significant class imbalance. We used the data before 2016 to construct our training data. Following the design of Task 2 in the manuscript, we randomly generated 10 training samples, each comprising 4000 positive examples and 4000 negative examples. The dataset was tested in two ways: balanced and unbalanced. For the balanced data, we randomly generated 10 testing samples, each consisting of 1734 positive examples and 1734 negative examples. For the unbalanced data, the entire set of 1734 positive cases and 71,740 negative cases was used as the test set. Similarly, we adopted XGBoost, which has the best performance, to compare the performance of different models. The results show that our proposed model can significantly outperform the baseline method (3Cs model) and that each dimension provides a critical contribution to idea screening.

Table XIV
Evaluation results of the test (XGBoost).

	XGBoost: Data-Split date is May. 1st, 2020									
	Balanced Testing Dataset (10 samples) (Implemented ideas = 1734) (Unimplemented ideas = 1734)					Unbalanced Testing Dataset (Implemented ideas = 1734) (Unimplemented ideas = 71,740)				
	Baseline	Baseline + D3	Baseline + D2	Baseline + D1	Baseline + D1 + D2 + D3	Baseline	Baseline + D3	Baseline + D2	Baseline + D1	Baseline + D1 + D2 + D3
Acc.	0.616 (0.027)	0.659*** (0.017)	0.644** (0.020)	0.661*** (0.023)	**0.683*** (0.016)**	0.631 (0.024)	0.652* (0.026)	0.649* (0.022)	0.670** (0.049)	**0.687*** (0.025)**
Pre.	0.617 (0.027)	0.660*** (0.017)	0.646** (0.022)	0.665*** (0.026)	**0.681*** (0.015)**	0.512 (0.002)	0.517*** (0.002)	0.514** (0.002)	0.518*** (0.002)	**0.532*** (0.005)**
Rec.	0.616 (0.027)	0.659*** (0.017)	0.644** (0.020)	0.661*** (0.023)	**0.680*** (0.016)**	0.612 (0.025)	0.657*** (0.017)	0.641** (0.020)	0.659*** (0.024)	**0.686*** (0.012)**
F1	0.615 (0.027)	0.659*** (0.017)	0.643** (0.019)	0.660*** (0.023)	**0.679*** (0.016)**	0.422 (0.009)	0.436*** (0.010)	0.426 (0.008)	0.444*** (0.019)	**0.469*** (0.011)**
AUC	0.641 (0.042)	0.711*** (0.026)	0.713*** (0.020)	0.725*** (0.035)	**0.754*** (0.022)**	0.639 (0.039)	0.708*** (0.026)	0.710*** (0.025)	0.724*** (0.034)	**0.774*** (0.012)**
SRC	0.244 (0.072)	0.365*** (0.045)	0.369*** (0.034)	0.390*** (0.060)	**0.433*** (0.039)**	0.075 (0.021)	0.113*** (0.014)	0.114*** (0.014)	0.121*** (0.018)	**0.151*** (0.008)**
NDCG	0.742 (0.054)	0.817*** (0.024)	0.851*** (0.020)	0.864*** (0.031)	**0.889*** (0.022)**	0.126 (0.116)	0.142 (0.091)	0.147 (0.070)	0.226** (0.088)	**0.352*** (0.113)**

Note 1. Pre. = precision; Rec. = recall; F1 = macro F1 score; Acc. = accuracy.
Note 2. Significant results compared with the baseline model are highlighted in bold, where * means p < 0.1; ** means $p < 0.05$; and *** means $p < 0.01$.
Note 3. The number in parentheses is the standard deviation of model performance in terms of the given metric.
Note 4. D1 is the exploration and exploitation of digital resources; D2 is the exploration and exploitation of direct interactions; and D3 is the exploration and exploitation of ideas and their comments.

To further evaluate the generalizability of the EEVC-based idea screening model, we collected a dataset from the Huawei Product Design Forum, comprising 20,064 idea entries submitted between December 2015 and January 2023. Among them, 7422 were labeled positive cases, and 12,642 were labeled negative cases. Following the evaluation procedure of Task 1, we conducted 10 rounds of random sampling, with each sample consisting of 7000 positive and 7000 negative instances. For each sample, we performed 10-fold cross-validation.

The results, summarized in Table XV, demonstrate that the EEVC model consistently outperformed all the baseline models. Moreover, each incremental component added to the baseline (3Cs model) resulted in a significant performance improvement. Under the XGBoost classifier, the EEVC model achieved performance gains over the baseline of 4.6 %, 4.3 %, 4.6 %, 4.3 %, 4.3 %, and 6.4 % in terms of precision, recall, the F1 score, accuracy, the AUC, and the SRC, respectively.

These findings suggest that the EEVC model has strong predictive power across different user cocreation platforms, highlighting its robust generalizability.

Table XV
Evaluation results of new data from the Huawei forum.

Cross-validation	Random Forest						XGBoost					
Feature Sets	Pre.	Rec.	F1	Acc.	AUC	SRC	Pre.	Rec.	F1	Acc.	AUC	SRC
Baseline	0.857 (0.013)	0.835 (0.015)	0.846 (0.009)	0.848 (0.009)	0.911 (0.008)	0.721 (0.014)	0.863 (0.012)	0.852 (0.014)	0.857 (0.009)	0.858 (0.009)	0.924 (0.007)	0.737 (0.012)

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Table XV (*continued*)

Cross-validation	Random Forest						XGBoost					
Feature Sets	Pre.	Rec.	F1	Acc.	AUC	SRC	Pre.	Rec.	F1	Acc.	AUC	SRC
Baseline + D3	0.885*** (0.012)	0.856*** (0.014)	0.870*** (0.009)	0.872*** (0.009)	0.941*** (0.006)	0.763*** (0.010)	0.880*** (0.011)	0.870*** (0.013)	0.875*** (0.008)	0.875*** (0.008)	0.942*** (0.006)	0.766*** (0.010)
Baseline + D2	0.895*** (0.010)	0.871*** (0.013)	0.883*** (0.008)	0.885*** (0.008)	0.950*** (0.005)	0.781*** (0.008)	0.885*** (0.011)	0.875*** (0.013)	0.880*** (0.008)	0.880*** (0.008)	0.947*** (0.005)	0.774*** (0.009)
Baseline + D1	0.895*** (0.011)	0.867*** (0.013)	0.881*** (0.008)	0.883*** (0.008)	0.950*** (0.005)	0.779*** (0.009)	0.893*** (0.010)	0.881*** (0.013)	0.887*** (0.008)	0.888*** (0.008)	0.952*** (0.005)	0.784*** (0.009)
Baseline + D1 + D2 + D3	0.910*** (0.011)	0.887*** (0.012)	0.898*** (0.009)	0.901*** (0.008)	0.965*** (0.005)	0.798*** (0.008)	0.909*** (0.011)	0.895*** (0.012)	0.903*** (0.008)	0.901*** (0.008)	0.967*** (0.005)	0.801*** (0.008)

Note 1. Pre. = precision; Rec. = recall; F1=F1 score; Acc. = accuracy.
Note 2. Significant results compared with the baseline model are highlighted in bold, where * indicates p < 0.1; ** indicates $p < 0.05$; and *** indicates $p < 0.01$.
Note 3. The number in parentheses is the standard deviation of model performance in terms of the given metric.
Note 4. D1 is the exploration and exploitation of digital resources; D2 is the exploration and exploitation of direct interactions; and D3 is the exploration and exploitation of ideas and their comments.

Data availability

Data will be made available on request.

References

[1] V. Krishnan, C.H. Loch, A retrospective look at production and operations management articles on new product development, Prod. Oper. Manag. 14 (4) (2005) 433–441.
[2] A. Simaei, R. Hirschheim, H. Schneider, Idea crowdsourcing platforms for new product development: a study of idea quality and the number of submitted ideas, Decis. Support. Syst. 175 (2023) 114041.
[3] C. Billington, R. Davidson, Leveraging open innovation using intermediary networks, Prod. Oper. Manag. 22 (6) (2013) 1464–1477.
[4] S. Hoornaert, M. Ballings, E.C. Malthouse, D. Van den Poel, Identifying new product ideas: waiting for the wisdom of the crowd or screening ideas in real time, J. Prod. Innov. Manag. 34 (5) (2017) 580–597.
[5] L.J. Kornish, J. Hutchison-Krupat, Research on idea generation and selection: implications for management of technology, Prod. Oper. Manag. 26 (4) (2017) 633–651.
[6] J. Goldenberg, D.R. Lehmann, D. Mazursky, The idea itself and the circumstances of its emergence as predictors of new product success, Manag. Sci. 47 (1) (2001) 69–84.
[7] D. Arnott, Cognitive biases and decision support systems development: a design science approach, Inf. Syst. J. 16 (1) (2006) 55–78.
[8] E.H. Hwang, P.V. Singh, L. Argote, Jack of all, master of some: information network and innovation in crowdsourcing communities, Inf. Syst. Res. 30 (2) (2019) 389–410.
[9] N. Jia, X.M. Luo, Z. Fang, C.C. Liao, When and how artificial intelligence augments employee creativity, Acad. Manag. 67 (1) (2024) 5–32.
[10] J.J. Bell, C. Pescher, G.J. Tellis, J. Fuller, Can AI help in ideation? A theory-based model for idea screening in crowdsourcing contests, Mark. Sci. 43 (1) (2023) 54–72.
[11] M. Rhyn, I. Blohm, A machine learning approach for classifying textual data in crowdsourcing, in: Proceedings of the 13th International Conference on Wirtschaftsinformatik (WI), 2017, pp. 1–15.
[12] C.R. Schaffhausen, T.M. Kowalewski, Large scale needs-based open innovation via automated semantic textual similarity analysis, in: Proceedings of the International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, 2015, pp. 1–11.
[13] B.L. Bayus, Crowdsourcing new product ideas over time: an analysis of the Dell IdeaStorm community, Manag. Sci. 59 (1) (2013) 226–244.
[14] M. Li, A. Kankanhalli, S.H. Kim, Which ideas are more likely to be implemented in online user innovation communities? An empirical analysis, Dec. Support Syst. 84 (2016) 28–40.
[15] I. Lykourentzou, F. Ahmed, C. Papastathis, I. Sadien, K. Papangelis, When crowds give you lemons: filtering innovative ideas using a diverse-bag-of-lemons strategy, Proc. ACM Human Comput. Interact. 2 (2018) 1–23.
[16] P.M. Di Gangi, M. Wasko, Steal my idea! Organizational adoption of user innovations from a user innovation community: a case study of Dell IdeaStorm, Dec. Support Syst. 48 (1) (2009) 303–312.
[17] H. He, Y. Ma, Imbalanced Learning: Foundations, Algorithms, and Applications, Wiley-IEEE Press, New York, 2013.
[18] C. Grönroos, Conceptualising value co-creation: a journey to the 1970s and back to the future, J. Mark. Manag. 28 (13–14) (2012) 1520–1534.
[19] A. Majchrzak, A. Malhotra, Effect of knowledge-sharing trajectories on innovative outcomes in temporary online crowds, Inf. Syst. Res. 27 (4) (2016) 685–703.
[20] J.G. March, Exploration and exploitation in organizational learning, Organ. Sci. 2 (1) (1991) 71–87.
[21] V. Aggarwal, E.H. Hwang, Y. Tan, Learning to be creative: a mutually exciting spatiotemporal point process model for idea generation in open innovation, Inf. Syst. Res. 32 (4) (2021) 1214–1235.
[22] M.K. Poetz, M. Schreier, The value of crowdsourcing: can users really compete with professionals in generating new product ideas? J. Prod. Innov. Manag. 29 (2) (2012) 245–256.
[23] A.R. Hevner, S.T. March, J. Park, S. Ram, Design science in information systems research, MIS Q. 28 (1) (2004) 75–105.
[24] J.G. Walls, G.R. Widmeyer, O.A. El Sawy, Building an information system design theory for vigilant EIS, Inf. Syst. Res. 3 (1) (1992) 36–59.
[25] Q. Liu, Q. Du, Y. Hong, W. Fan, S. Wu, User idea implementation in open innovation communities: evidence from a new product development crowdsourcing community, Inf. Syst. J. 30 (5) (2020) 899–927.
[26] C. Riedl, V.P. Seidel, Learning from mixed signals in online innovation communities, Organ. Sci. 29 (6) (2018) 1010–1032.
[27] S. Erat, Making the best idea better: the role of idea pool structure, Prod. Oper. Manag. 26 (10) (2017) 1946–1959.
[28] K. Eling, A. Griffin, F. Langerak, Consistency matters in formally selecting incremental and radical new product ideas for advancement, J. Prod. Innov. Manag. 33 (2016) 20–33.
[29] M.J. Van Damme, F. Anseel, W. Duyck, E.F. Rietzschel, Strategies to improve selection of creative ideas: an experimental test of epistemic and social motivation in groups, Creat. Innov. Manag. 28 (1) (2019) 61–71.
[30] L.J. Kornish, S.M. Jones, Raw ideas in the fuzzy front end: verbosity increases perceived creativity, Mark. Sci. 40 (6) (2021) 1106–1122.
[31] L.J. Kornish, K.T. Ulrich, Opportunity spaces in innovation: empirical analysis of large samples of ideas, Manag. Sci. 57 (1) (2011) 107–128.
[32] Y. Huang, P. Vir Singh, K. Srinivasan, Crowdsourcing new product ideas under consumer learning, Manag. Sci. 60 (9) (2014) 2138–2159.
[33] X. Liu, G.A. Wang, W. Fan, Z. Zhang, Finding useful solutions in online knowledge communities: a theory-driven design and multilevel analysis, Inf. Syst. Res. 31 (3) (2020) 731–752.
[34] W.Y. Ayele, G. Juell-Skielse, A process model for generating and evaluating ideas: the use of machine learning and visual analytics to support idea mining, in: Electronic Government and the Information Systems Perspective: 9th International Conference, EGOVIS, 2020, pp. 189–203.
[35] S.M. Lundberg, S.I. Lee, A unified approach to interpreting model predictions, Adv. Neural Inf. Proces. Syst. 30 (2017) 4766–4775.
[36] K. Wang, B. Dong, J. Ma, Towards computational assessment of idea novelty, in: Proceedings of the 52nd Hawaii International Conference on System Sciences, Grand Wailea, Maui, HI, USA, January 8–11, 2019.
[37] X. Lin, B. Shao, X. Wang, Employees' perceptions of chatbots in B2B marketing: affordances vs. disaffordances, Ind. Mark. Manag. 101 (2022) 45–56.
[38] M. Beretta, Idea selection in web-enabled ideation systems, J. Prod. Innov. Manag. 36 (1) (2019) 5–23.
[39] J. Ma, Y. Lu, S. Gupta, User innovation evaluation: empirical evidence from an online game community, Decis. Support. Syst. 117 (2019) 113–123.
[40] M. Klein, A.C.B. Garcia, High-speed idea filtering with the bag of lemons, Decis. Support. Syst. 78 (2015) 39–50.
[41] G.C. Kane, M. Alavi, Information technology and organizational learning: an investigation of exploration and exploitation processes, Organ. Sci. 18 (5) (2007) 796–812.
[42] A. Ferecatu, A. De Bruyn, Understanding managers' trade-offs between exploration and exploitation, Mark. Sci. 41 (1) (2022) 139–165.
[43] C. Andriopoulos, M.W. Lewis, Exploitation-exploration tensions and organizational ambidexterity: managing paradoxes of innovation, Organ. Sci. 20 (4) (2009) 696–717.
[44] A. Zadeh, A. Jeyaraj, A multistate modeling approach for organizational cybersecurity exploration and exploitation, Decis. Support. Syst. 162 (2022) 113849.
[45] R. Azoulay-Schwartz, S. Kraus, J. Wilkenfeld, Exploitation vs. exploration: choosing a supplier in an environment of incomplete information, Decis. Support. Syst. 38 (1) (2004) 1–18.

[46] H. Liang, N. Wang, Y. Xue, Juggling information technology (IT) exploration and exploitation: a proportional balance view of IT ambidexterity, Inf. Syst. Res. 33 (4) (2022) 1386–1402.
[47] S. Djelassi, I. Decoopman, Customers' participation in product development through crowdsourcing: issues and implications, Ind. Mark. Manag. 42 (5) (2013) 683–692.
[48] D. Schlagwein, N. Bjørn-Andersen, Organizational learning with crowdsourcing: the revelatory case of LEGO, J. Assoc. Inf. Syst. 15 (11) (2014) 754–778.
[49] K.W. Chan, S.Y. Li, J. Ni, J.J. Zhu, What feedback matters? The role of experience in motivating crowdsourcing innovation, Prod. Operat. Manag. 30 (1) (2021) 103–126.
[50] Q. Liu, Z. Yang, X. Cai, Q. Du, W. Fan, The more, the better? The effect of feedback and user's past successes on idea implementation in open innovation communities, J. Assoc. Inform. Sci. Technol. 73 (3) (2022) 376–392.
[51] J.O. Wooten, K.T. Ulrich, Idea generation and the role of feedback: evidence from field experiments with innovation tournaments, Prod. Oper. Manag. 26 (1) (2017) 80–99.
[52] R. Hanna, S. Mullainathan, J. Schwartzstein, Learning through noticing: theory and evidence from a field experiment, Q. J. Econ. 129 (3) (2014) 1311–1353.
[53] L. Fleming, Recombinant uncertainty in technological search, Manag. Sci. 47 (1) (2001) 117–132.
[54] H. Casakin, Visual analogy as a cognitive strategy in the design process. Expert versus novice performance, J. Des. Res. 4 (2) (2004) 197–217.
[55] V. Pestov, Is the k-nn classifer in high dimensions affected by thecurse of dimensionality? Comput. Math. Appl. 65 (2013) 1427–1437 (Grasping Complexity).
[56] R. Cross, L. Sproull, More than an answer: information relationships for actionable knowledge, Organ. Sci. 15 (4) (2004) 446–462.
[57] J.M. Berg, Balancing on the creative highwire: forecasting the success of novel ideas in organizations, Adm. Sci. Q. 61 (3) (2016) 433–468.
[58] S. Chung, A. Animesh, K. Han, A. Pinsonneault, Software patents and firm value: a real options perspective on the role of innovation orientation and environmental uncertainty, Inf. Syst. Res. 30 (3) (2019) 1073–1097.
[59] D.M. Blei, A.Y. Ng, M.I. Jordan, J. Lafferty, Latent Dirichlet allocation, J. Mach. Learn. Res. 3 (2003) 933–1022.
[60] Q.V. Le, T. Mikolov, Distributed representations of sentences and documents, Int. Conf. Mach. Learn. (2014) 1188–1196.
[61] N. Reimers, I. Gurevych, Sentence-BERT: sentence embeddings using siamese BERT-networks, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), 2019.
[62] T.L. Griffiths, M. Steyvers, Finding scientific topics, Proc. Natl. Acad. Sci. USA 101 (Supplement 1) (2004) 5228–5235.
[63] Y.R. Tausczik, J.W. Pennebaker, The psychological meaning of words: LIWC and computerized text analysis methods, J. Lang. Soc. Psychol. 29 (1) (2009) 24–54.
[64] J. Uotila, M. Maula, T. Keil, S.A. Zahra, Exploration, exploitation, and financial performance: analysis of S&P 500 corporations, Strateg. Manag. J. 30 (2) (2009) 221–231.
[65] N. Japkowicz, S. Stephen, The class imbalance problem: a systematic study, Intell. Data Anal. 6 (5) (2002) 429–449.
[66] A. Fernández, S. García, M. Galar, R.C. Prati, B. Krawczyk, F. Herrera, Learning from imbalanced data sets, in: Data Intrinsic Characteristics, Springer, Cham, 2018, pp. 253–277.
[67] G.E. Batista, R.C. Prati, M.C. Monard, A study of the behavior of several methods for balancing machine learning training data, ACM SIGKDD Explor. Newslett. 6 (1) (2004) 20–29.
[68] M. Qin, S. Li, F. Cai, W. Zhu, S. Qiu, Where are your ideas going? Idea adoption in online user innovation communities, Eur. J. Innov. Manag. 27 (2024) 2122–2148.
[69] L. Dahlander, M. Beretta, A. Thomas, S. Kazemi, M.H. Fenger, L. Frederiksen, Weeding out or picking winners in open innovation? Factors driving multi-stage crowd selection on LEGO ideas, Res. Policy 52 (2023) 104875.
[70] C.Y. Hsiang, J.T. Rayz, Predicting popular contributors in innovation crowds: the case of my Starbucks ideas, Inf. Technol. People 35 (2022) 494–509.
[71] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Comput. 9 (8) (1997) 1735–1780.
[72] A. Rai, Explainable AI: from black box to glass box, J. Acad. Mark. Sci. 48 (1) (2020) 137–141.
[73] G. Chandrashekar, F. Sahin, A survey on feature selection methods, Comput. Electr. Eng. 40 (1) (2014) 16–28.
[74] N. Gorban, I.Y. Tyukin, Blessing of dimensionality: mathe-matical foundations of the statistical physics of data, Philos. Trans. R. Soc. A Math. Phys. Eng. Sci. 376 (2018) 20170237.
[75] A. Camuffo, A. Cordova, A. Gambardella, C. Spina, A scientific approach to entrepreneurial decision making: evidence from a randomized control trial, Manag. Sci. 66 (2) (2020) 564–586.
[76] P. Criscuolo, L. Dahlander, T. Grohsjean, A. Salter, The sequence effect in panel decisions: evidence from the evaluation of research and development projects, Organ. Sci. 32 (4) (2021) 987–1008.
[77] D. Lee, K. Hosanagar, How do recommender systems affect sales diversity? A cross-category investigation via randomized field experiment, Inform. Syst. Res. 30 (1) (2019) 239–259.
[78] L.W. Ku, H.H. Chen, Mining opinions from the Web: beyond relevance retrieval, J. Am. Soc. Inf. Sci. Technol. 58 (2007) 1838–1850.
[79] D. Hasler, S.E. Suesstrunk, Measuring colorfulness in natural images, in: Human Vision and Electronic Imaging VIII 5007, SPIE, June 2003, pp. 87–95.

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